History of Changes

Version	Publication Date	Description of Changes
1.0	30/08/2022	Annex 1 Part B Initial version with the following updates: Updating of the shortname of partner originally referred to as AGC to their official shortname, which is INTERPANE. These changes have been made across the Part B everywhere where this Partner was mentioned, as well as in all the diagrams. The Gantt on Page 34 has been updated according to the revised Deliverables in Part A of the DoA (whereby any deliverable that had incorporated numerous versions has been attributed a separate deliverable per each version date).
1.1	12/09/2022	In order to address the following comment in the ESR: 'The proposal indicates several barriers to outcomes and impacts and mitigation measures are determined properly. However, some regulatory issues related to validation of the proposed nanocoatings are not addressed to a full extent', we have added the following statement in Table 15 on page 24: 'Regulatory issues related to validation of the proposed nanocoatings will be duly assessed during via the cocreation work that will be performed with industry in Task 4.1 Cocreation for an application-oriented development, as well as in WP5- Validation and Impact. We have a regulatory partner (UNE) in our consortium: their role is to address regulatory issues related to validation of the proposed nanocoatings.' In order to address the following comment in the ESR in relation to the 'Do no significant harm principle', whereby the experts stated 'Given the particular character of (some) materials proposed to be used as components of the new anti-pathogen coatings, potential harm to humans and ecosystems cannot be fully excluded. There is no sufficient information provided at this stage to assess the full compliance of the proposal with "Do no significant harm" principle.', we have added the following statement on page 17: 'However, we will continually monitor this to ensure full compliance, and in particular this will be assessed in Task 2.3 Toxicity & ecotoxicity. We will also leverage the expertise of regulatory partner UNE.' The Shortname of INTERPANE has been updated in Table 3.1h: 'Purchase costs' items (travel and subsistence, equipment and other goods, works and services) and 'Other costs categories' items (e.g. internally invoiced goods and services) on page 36.
1.2	04/10/2022	Table 3.1h: 'Purchase costs' items (travel and subsistence, equipment and other goods, works and services) and 'Other costs categories' items has been modified to include audit costs (CFS- Certificate on the Financial Statements) for Partners SIEON and INTERPANE on page 36.
1.3	12/10/2022	In this version of the DoA, while keeping the original deliverables, we provide a greater description as to why we split some deliverables during the Contract Negotiations process. The sole reason was to ensure that any deliverable that originally (at proposal stage) had multiple dates of delivery, would be split into a separate deliverable for each date of delivery (and the deliverables were then numbered (ordered) according to the date of delivery). For example, at proposal stage, we originally had *D1.3 Report on wet chemistry & biobased coatings* to be delivered at M12 and M24. During the preparation of the DoA we split this deliverable in two, to ensure that each date of delivery would have its own Deliverable, so that each version could be uploaded in its unique placeholder in the electronic system. As an example: the current status in the DoA is that we have *D1.3 1st Report on wet chemistry & biobased coatings* to be delivered at M12, and *D1.4 2nd Report on wet chemistry & biobased coatings* to be delivered at M24. Likewise, we performed the same process for all deliverables that at proposal stage had multiple dates of delivery for the same deliverable.
1.4	30/06/2025	Annex 1 Part B Request of amendment

		Pag 38 New gantt due to the 6 months requested extension **WP1:** End month shifts from M27 to M36: - Task 1.4 end month shifted from M27 to M36 - no changes for the other tasks *Deliverables* - *D1.6 2nd Report on coating optimization* New deliverables at M36 *Milestones* - *MS10* new due date at M36 **WP2:** End month shifts from M30 to M41 - Task 2.1 end month shifted from M24 to M36. - Task 2.2 end month shifted from M27 to M41. - Task 2.3 end month shifted from M27 to M41. - Task 2.4 end month shifted from M30 to M41. *Deliverable* - *D2.7 3rd Technical periodical report describing the antiviral activity of the coatings* New deliverables at M41 - *D2.8 3rd Technical periodical report describing cytotoxicity* New deliverables at M41 - *D2.6* new due date at M41 *Milestones* - *MS8* new due date at M41 - *MS9* new due date at M41 - *MS12* new due date at M41 **WP3:** end month shifts from M27 to M41 - Task 3.1 end month shifted from M27 to M41. - Task 3.2 end month shifted from M27 to M41. - Task 3.3 end month shifted from M27 to M41. *Deliverable* - *D3.7 3rd Report on physical-chemical characterization of coatings* New deliverables at M41 - *D3.8 3rd Report on the mechanical characterization of the coatings* New deliverables at M41 - *D3.9 3rd Report on chemical resistance & ageing of the coatings* New deliverables at M41 *Milestones* - *MS11* new due date at M41 - *MS13* new due date at M41 **WP4:** end month shifts from M32 to M42 - Task 4.3 end month shifted from M30 to M42. - Task 4.4 end month shifted from M32 to M42. *Deliverable* - *D4.10* new due date at M42 - *D4.11* new due date at M42 **WP5:** end month shifts from M36 to M42 - Task 5.2 end month shifted from M36 to M42. - Task 5.3 end month shifted from M36 to M42. *Deliverable* - *D5.2* new due date at M42 - *D5.3* new due date at M42 *Milestones* - *MS14* new due date at M36 - *MS15* new due date at M42

		WP6: end month shifts from M36 to M42 (all the tasks) *Deliverable* - *D6.7* new due date at M42 - *D6.8* new due date at M42 - *D6.9* new due date at M42 **WP7:** end month shifts from M36 to M42 - Task 5.1 end month shifted from M36 to M42. - Task 5.2 end month shifted from M36 to M42. - Task 5.3 end month shifted from M36 to M42. *Deliverable* - *D7.4* new due date at M42 **Pag 39-41** New table 3.1h 'Purchase costs' items (travel and subsistence, equipment and other goods, works and services) and 'Other costs categories' items (e.g. internally invoiced goods and services) **1 - POLITO** **- € 11.700 from C.2 Equipment to C.3 Other goods, works and services** **- € 4.300 from C.2 Equipment to C.1 Travel** **Update budget** **C.1 Travel 21300 €** **C.2 Equipment 4000 €** **C.3 Other goods, works and services 72600€** *With reference to the budget item "Equipment", POLITO asks to allocate part of the initial amount allocated in the 'ESTIMATED BUDGET FOR THE ACTION' to Other goods, works and services. We have organized the tests in order to use the equipment already present in the department laboratory and in the interdepartmental J-Tech laboratory of the Politecnico di Torino, making it unnecessary to purchase additional equipment. The budget reallocated to item C.3 Other goods, works and services was used to stipulate a temporary services contract with a technician specialized in the development and characterization of coating via co-sputtering in order to support also the upscaling of the process.* *In addition, given the interest generated by the project results, the invitations to participate in dissemination events, as conference with invited presentations, were higher than expected.* **3 - UNITO** **- € 25.000,00 from C.2 Equipment costs category to A.1 Personnel costs category** **Update budget** **C.2 Equipment 25000 €** *With reference to the budget item "Equipment", UNITO asks to remodulate the initial amount allocated in the 'ESTIMATED BUDGET FOR THE ACTION'. It is proposed to reduce Equipment expenditure item by €25,000 and relocate it to item "A. Personnel costs", resulting in a total increase of 12 person-months in the research group's effort (from 58 to 70). This request stems from the need to strengthen the group's component dedicated to laboratory activities, which hold a significant weight, through the ad hoc recruitment of a young researcher. In addition, an internal simulation has revealed that the project will incur in a maximum accounting depreciation of €25,000 for the equipment needed to implement it (all acquired during the first two semesters of the project).* *These above-mentioned requests do not imply substantial changes to the research activities of the project* **4 - FAU** **- 5.000 € from A.1 Personnel cost to C.3 Other goods.** **- 6.000 € from C.1 Travel to C.3 Other goods.** **Update budget** **C.1 Travel 9000 €** **C.3 Other goods 28600 €** *FAU asks to relocate 5.000 € from the initial budget allocated to the category "A.1 Personnel cost" to "C.3 Other goods". The remaining budget in the category "A.1 Personnel cost" is sufficient to cover the personnel costs of the contract(s) (man-months) until end of*

		the project extension period (M42). Furthermore, FAU asks to relocate 6.000 € from the initial budget allocated to the category "C.1 Travel" to "C.3 Other goods". The remaining budget in the category "C.1 Travel" is sufficient to cover all pending reimbursements and futures trips until end of the project extension period (M42). The modification of the budget to the category "C.3 Other goods" is required considering the extensive research work and the variety of coatings produced until M30 of the project. The contribution of FAU to the project in its initial stage was mainly focused on the preparation, optimization and preliminary characterization of the coatings, all material and resources intensive tasks, which necessitated more funds to obtain the lab consumables and testing kits/techniques. The relocated budget for "other goods" will be used to continue the optimization of the most promising coating types. **5 - GV FILTRI** **- 5.000 € from C.1 Travel to A.1 Personnel cost.** **- 30.000 € from C.3 Other goods to A.1 Personnel cost** **Update budget** **C.1 Travel 10000 €** **C.3 Other goods 40000,00 €** *The budget reallocation has become necessary due to the variety of product types and the different coatings applied, which have introduced a higher level of complexity than initially anticipated. The search for applicable standards and reference guidelines required more person-hours than originally planned. In addition, durability tests were conducted over longer timeframes and on substrates that had not been foreseen, with suboptimal results. As a result, it was necessary to produce new samples to be coated and tested again, further increasing the workload. As the leader of WP5, these activities have entailed—and will continue to entail—a greater effort in managing the work package and coordinating the partners involved.* **7 - PRIREV** **€ 50,375 from C.2 Equipment to C.3 Other goods** **Update budget** **C.2 Equipment 68800€** **C.3 Other goods 144375,00 €** *For external characterization services provided by laboratories equipped with advanced and more suitable instrumentation for the evaluation of coating properties.* *This reallocation reflects the strategic shift we adopted during the project implementation, prioritizing access to high-performance external facilities over in-house equipment acquisition. These services have been crucial for the technical development and validation of our coatings, allowing for more accurate, efficient, and reliable characterization. In particular, a significant portion of the external services are based on X-ray Fluorescence (XRF) analysis, which has proven essential to support our work on PVD coatings with complex multilayer structures and variable compositions, including the incorporation of silver at different concentrations. The XRF technique provides several key advantages:* *•Precise elemental composition analysis: It enables the identification and quantification of elements in the coatings, ensuring that the intended material composition—including silver content—is correctly achieved.* *•Verification of layer uniformity: XRF allows us to check the homogeneity of the coatings across the surface, which is critical for process validation and quality assurance.* *•Multilayer structure insight: Although not a direct thickness measurement technique, XRF helps in detecting variations in the elemental distribution across multilayer structures, supporting the fine-tuning of our deposition processes.* *•Non-destructive and rapid: The non-destructive nature of XRF, combined with fast turnaround and low cost, makes it an ideal tool for regular analysis during the optimization and prototype production phases. Therefore, the use of XRF-equipped external labs has been strategically important to meet the project's technical objectives and ensure high-quality, reproducible coatings.* *To justify these costs, a service contract covering the execution of the external tests during the project period is fundamental. This contract outlines the scope of work, including the number and type of analyses to be performed, and can be used to classify the related expenses as "services" rather than subcontracting, in line with the criteria you outlined.*

		8 - IRIS **Just modifying the description in the other goods and services:** *current: Analysers - 60 k; Optomechanical materials - 10k; computing for prototype (5k); software licenses (4k); Audit certificate (4k); IP cost (Patent lawyer & patent application) - 8k; subscription to events & exhibition space - 6k; Data Management Cost (6k)* *updated: Analysers - 50 k; QT SOFTWARE LICENSE FOR THE DEVELOPMENT OF THE APPLICATIONS USED IN THE MONITORING PROTOTYPE - 10K€ Optomechanical materials - 10k; computing for prototype (5k); software licenses (4k); Audit certificate (4k); IP cost (Patent lawyer & patent application) - 8k; subscription to events & exhibition space - 6k; Data Management Cost (6k)*

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1. Excellence

1.1 Objectives and ambition

Introduction: The respiratory illness coronavirus disease 2019 (COVID-19) that has gripped the world in a global health and economic crisis since early 2020, is caused by an enveloped RNA virus known as severe acute respiratory syndrome coronavirus 2 (SARS-CoV-2). Despite strict lockdowns and confinement measures that were quickly put in place, the disease has spread across the globe; to date World Health Organization (WHO) reports > 228 million confirmed cases of COVID-19 and nearly 4.7 million deaths. The enduring pandemic has thrown the global economy into its worst economic crisis since World War II, with crippling impacts on healthcare and socioeconomic costs.

Potential routes of infection according to the WHO are droplet, contact, fomite (contaminated surfaces, whereby SARSCoV-2 remains stable on these surfaces for many hours) and airborne (aerosol) transmissions. **In the case of aerosol transmission, the air filtration and disinfection efficiency of air filtration/circulation units plays an important role in minimizing the virus transmission, especially in areas with limited external air supply** (e.g., public transportation vehicles) or in clinics and nosocomial settings[1]. On the other hand, in the case of fomite transmissions, **laboratory studies have discovered that the SARS-CoV-2 remains active on soft plastic materials** up to 72 h[2], and on the surfaces of metals, plastics, cottons, or surgical masks for tens of hours to even 7 days[3]. Beyond laboratory evidences, **studies in controlled hospital environments[4] and common environments, showed that SARS-CoV-2 with infectious capacity can be found in furniture elements in the environments of infected people**. Beyond SARS-CoV-2, **other viruses, such as norovirus, have proven their ability to transmit by fomites**[5] and the WHO estimates that millions of patients each year around the world are affected by nosocomial infections (mostly bacterial) while receiving healthcare in hospitals. **It is therefore vital to improve the protection for our healthcare systems, including workers and patients.**

The race to reduce the spread of the virus has catalysed research and development in the field of antiviral nanocoating developments, which has seen **a wave of antimicrobial/virucidal coatings and coated products entering the market** making numerous claims in relation to effectiveness and durability. However, the reality is that current products are marketed **without a robust science-based evidence** in relation to their real efficiency, durability in relevant environments for the applications or other effects such as toxicity; at best, **testing is based on current standards that cannot clearly define or guarantee the claimed properties of the products**. Even when supported by standard tests, some commercial claims are obscure and doubtful as there is simply insufficient evidence to support some claims in relation to the functionality on applications.

In this context, NanoBloc's aim is to provide **reliable Antimicrobial, Antiviral, and Antifungal Nanocoatings**, by **sustainable synthesis methods**, suitable for application on all **surfaces and textiles**.

To do so, NanoBloc® is mobilising an **interdisciplinary (materials, coating processes, microbiology & AI) cocreation alliance between 4 world-class universities & institutes, and 5 leading industrial companies (3 SMEs & 2 large enterprises) who together will cocreate, develop and upscale (from TRL3 to TRL6) reliable 'Made in Europe' antimicrobial, antifungal and antiviral coatings manufactured with industrially scalable, versatile and green technologies suitable for application on a variety of different substrates**: filter materials (air filtration units, face masks), textiles (protective clothing, mattress covers, aprons, wallpaper), and also on diverse high-traffic solid surfaces (door knobs, handles, handrails, sanitaryware-taps, etc.).

Building on previous encouraging results obtained by the consortium partners, NanoBloc® coatings will **be tested in working conditions** for each envisaged application, with particular emphasis on their **health & environment** safety.

NanoBloc® and HORIZON-CL4-2021-RESILIENCE-01-20 **are a unique opportunity for Europe to take the lead in the development of reliable Antimicrobial, Antiviral, and Antifungal Nanocoatings** that respond to market needs in terms of efficiency, durability and sustainability/ecofriendly properties, and that are backed with claims that are validated by solid science-based evidence on antimicrobial and virucidal mechanisms of nanocoatings. In doing so we will help Europe to build resilience against future global health crises.

Figure 1 below presents a schematic overview of the R&D work that will be performed during the 36 months duration of the project to advance the technology readiness levels of our chosen coating technologies. It also maps the involvement of the consortium of research-industry partners and illustrates the products that will be the focus of

[1] Verdenelli, M.C. et al. (2003). Efficacy of antimicrobial filter treatments on microbial colonization of air panel filters. J. Appl. Microbiol. 2003, 94, 9–15
[2] Van Doremalen, N. et al. (2020) Aerosol and Surface Stability of SARS-CoV-2 as Compared with SARS-CoV-1. N. Engl. J. Med. 2020, 382, 1564–1567
[3] Chin A.W.H., Chu J.T.S., Perera M.R.A., Hui K.P.Y., Yen H., Chan M.C.W., Peiris M., Poon L.L.M. Lancet Microbe. 2020;1
[4] Liu, Jie, et al. "The duration of SARS-CoV-2 positive in the environments of quarantine rooms: a perspective analysis." Intl. Journal of Infectious Diseases (2021).
[5] Canales, Robert A., et al. "Modeling the role of fomites in a norovirus outbreak." Journal of occupational and environmental hygiene 16.1 (2019): 16-26.

this application-oriented R&D cocreation partnership. We will pursue an exciting series of **groundbreaking research lines** to develop industrially scalable, versatile and green technology in order to obtain innovative products that will be brought to market by our industry participants.

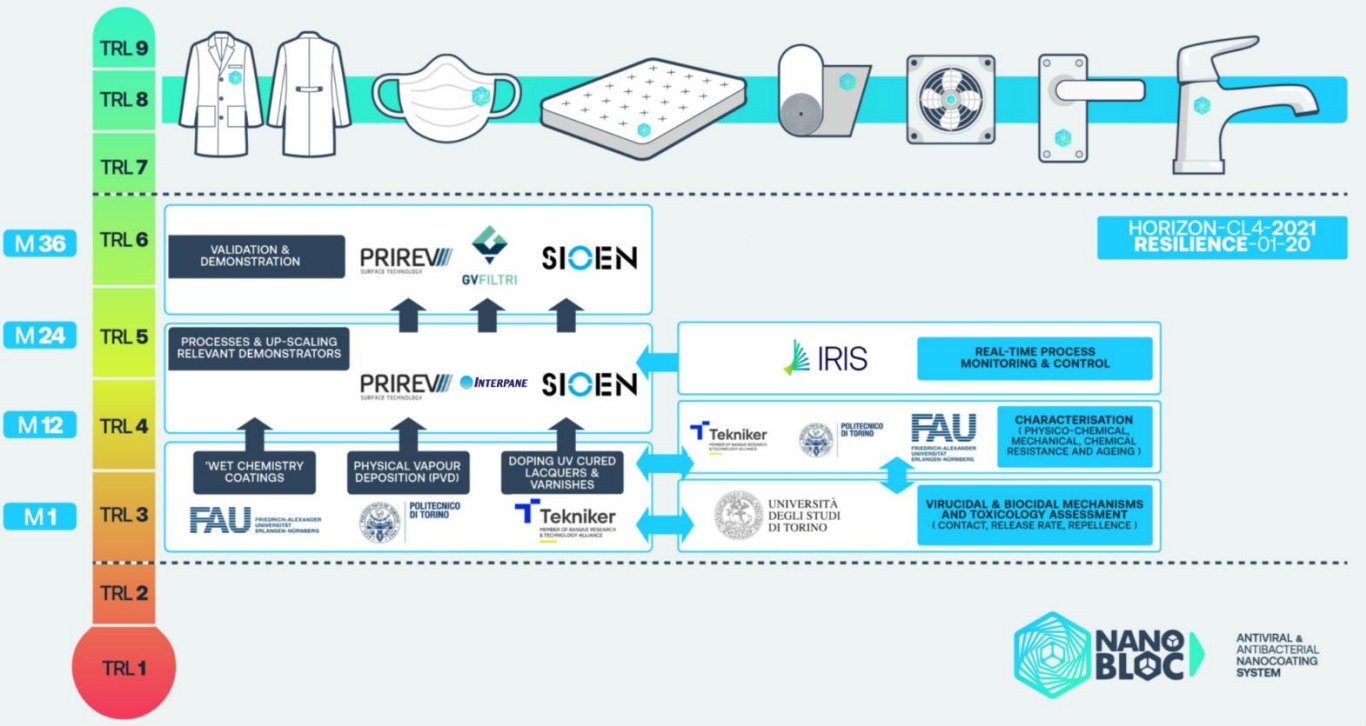
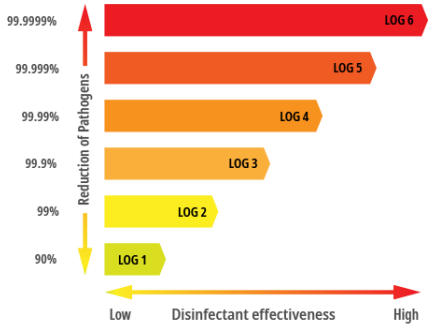


Figure 1- Schematical overview of the Nanobloc® Project

Measurable and verifiable objectives that are pertinent to the call topic and realistically achievable

Within the framework of this ambitious overall goal, the experienced and relevant consortium has codesigned NanoBloc establishing a series of 8 objectives and 5 measurable and verifiable Key Performance Indicators (KPIs) to be fulfilled within the project timeline (Table 1), and which will allow the long term expected outcomes also beyond Nanobloc project (discussed in Impact section).

Table 1- Project objectives and Key Performance Indicators

Specific Objectives (O)	Key Performance Indicators
O1-To develop new nanocoatings and processing routes: build on previous work in obtaining coatings effective against a range of pathogens (including virus & bacteria) using a novel and versatile platform of diverse nanocoating technologies- namely Physical Vapour deposition (PVD) nanocoatings (RF co-sputtering nanoAg/Silica, Sputtering & Cathodic arc of Ag doped nitrides, Cathodic arc, magnetron sputtering & HIPIMs for Cu doped nitrides and oxides), Organic/inorganic nanocomposite coatings (Nanoparticle doped UV lacquers), Wet Chemistry coatings (Electrophoretic deposition of biobased formulations, Sol-Gel hybrid coatings)-, enabling application on a variety of substrates relevant for different applications incl. metallic, plastic & textiles. **O2-To acquire a deeper understanding of the mechanisms of antiviral, antibacterial and antifungal effects:** to be able to reach O1 by tuning the coating properties to a) act fast against pathogens, b) have adequate durability for the application, and, c) avoid toxicological effects or environmentally harmful releases. **O3-To acquire and apply new knowledge to guarantee the durability and functionality of the coatings:** by characterizing structural, compositional and morphological features of the coatings and testing under different ageing and wear conditions to feedback the development of coatings and how ageing affects the functionality. **O4-To obtain scientific evidence of impact of the new Nanobloc coatings in relevant environments:** although many biocide coatings have proven their efficiency in laboratory testing, their real impact on	 **KPI1- High effectivity coatings against different pathogens in laboratory tests.** **Target: High reduction of a wide range of pathogens**, including virus (enveloped & not-enveloped), bacteria (Gram positive and negative) & fungi, with target values of 99.9999% reduction (log 6). **KPI2- High durability and safety of the coatings functionality.** **Target:** Biocide functionality in laboratory tests must remain at high

relevant environments has never been tested for coatings and has only been poorly addressed for Cu alloys[6]. It is therefore necessary to demonstrate the relevance of laboratory results in terms of impact on relevant environments. As such, NanoBloc will deploy a 1-year campaign to perform validation trials in relevant environments such as hospitals and shared areas in workplaces (toilets, meeting rooms…) to gather evidence (and open data) of the real effectivity of the coatings. **O5-To demonstrate the viability of industrialization and the application of the new NanoBloc coatings in relevant substrates & products:** by scaling up the selected processes and technologies in industry for the development of effective & durable biocidal/virucidal coatings. Contribution to generate new standards will support these industrial applications. **O6-To boost developments, reliability & sustainability of the processes 'by-design':** by using data-driven strategy enabled by Process Analytical Technology (PAT)-based real-time monitoring for obtaining critical material composition and process data, using Machine Learning techniques for predictive model control for process optimization, as well as for Life Cycle Analysis inventory to ensure the sustainability. **O7-To produce relevant demonstrators for different substrates & applications:** applying the new effective & durable biocidal/virucidal coatings on porous filter materials (air filtration units, face masks), textiles (protective clothing, mattress covers, aprons, wallpaper), on diverse high-traffic solid surfaces (doorknobs, handles, handrails, sanitaryware-taps). **O8-To validate the developments under relevant conditions & environments:** by assessing the performance of the demonstrators over a 12 month period by testing under standards for each application, as well as testing performance in application environments (San Luigo Hospital that hosts the UNITO laboratory, high traffic areas- air filtration units (POLITO, UNITO), Offices and toilets (TEKNIKER)-, to validate efficiency, durability, environmental friendliness and that there are no toxic effects for health & ecotoxicity impacts for the environment.	levels of reduction- 99.99% (log4)- after long-term exposure to relevant aging and wear conditions for the applications. **KPI3- Rapid action.** **Target:** Reduce the possibility of infection in between routine cleaning procedures and or increase the time of use of disposable personal protection equipment- reach 99.99% (log 4) reduction of pathogens in less than 15 minutes and 99.9999% (log 6) in 1 hour. **KPI4- Non-toxic and Ecotoxic effects of the coatings.** **Target:** Demonstrated toxicological safety for human health <30% cells dying on viability cell assays & non-ecotoxicity >50% survival of luminescent algae, following standards **KPI5- Proven effectivity and durability in relevant environments.** **Target:** >70% reduction in pathogen load vs. non-coated controls in surfaces in relevant environments while ensuring the durability of the effect over a 1-year long testing campaign. following successful lab-based weathering trials to guarantee their safe validation *in situ*.

Table 2 shows how these objectives respond to the call topic requests and how they are realistically achievable.

Table 2- Mapping of the call requests vs. the Nanobloc objectives and how they are realistically achievable

Pertinence with WP topic	How this is realistically achievable
'Sustainable synthesis of nanocoatings/nanocomposites (including bio-based materials) with effectiveness against a range of pathogens': **KPI1** High effectivity against different pathogens in laboratory tests **KPI3** Rapid action **KPI4** Non-toxic and Ecotoxic effects of the coatings **O1** To develop new nanocoatings and processing routes **O2** To acquire a deeper understanding of the mechanisms of antiviral & biocidal effects **O6** To boost development, reliability & sustainability of the processes	• One of the selected technologies for the synthesis of nanocoatings/ nanocomposites is Physical Vapour Deposition, considered one of the "**greenest**" and environmentally friendly coating technology, due to the lack of waste products to be disposed, to the "dry" process characteristics (no liquid involved) and to the low consumption of starting materials (coatings are few nanometres thick). • **All coatings proposed in NanoBloc® have been already demonstrated effective against several bacteria and fungi.** • POLITO's Ag/silica co-sputtered coating is effective on **viruses including respiratory syncytial virus (RSV), influenza virus A (FluVA)**[7]. The POLITO Ag/silica co-sputtered coating[8] & TEKNIKER Cu based coatings have already been tested & resulted **effective against SARS-CoV2** • Use new knowledge on **antiviral mechanisms** to select the most suitable technology for each application. • We will develop and/or scale up new synthesis **(including bio-based materials**) with a specific concern to process sustainability. • Up-scaling of the selected technologies in industry for the development of effective & durable biocidal/virucidal coatings will be facilitated by **Process Analytical Technology (PAT)-based real-**

[6] CHYDERIOTIS et al. New insights on antimicrobial efficacy of Cu surfaces in the healthcare environment: a systematic review. Clin Microbio Infec, 2018, 24 (11) 1130-1138.
[7] Nanostructured composite coating endowed with antiviral activity against human respiratory viruses deposited on fibre-based air filters. C. Balagna, R. Francese , S. Perero , D. Lembo, M. Ferraris, Surface & Coatings Technology 409 (2021) 12687. https://doi.org/10.1016/j.surfcoat.2021.126873
[8] Virucidal effect against coronavirus SARS-CoV-2 of a silver nanocluster/ silica composite sputtered coating
Cristina Balagna, Sergio Perero , Elena Percivalle, Edoardo Vecchio Nepita ,Monica Ferraris, Open Ceramics 1 (2020) 100006, https://doi.org/10.1016/j.oceram.2020.100006

	time monitoring & **Machine Learning for predictive model control for process optimization**
'Application methods (both on surfaces and textiles)': **O1** To develop new nanocoatings and processing routes **O5** To demonstrate the viability of industrialization and the application the new Nanobloc coatings in relevant substrates & products **O7 P**roduce relevant demonstrators for different substrates& applications	• All coatings proposed within NanoBloc® **have been already successfully applied to several surfaces and textiles.** • Thanks to the participation of a **very strong and relevant industrial value chain,** we will be able to apply coatings on surfaces and textiles thanks to the expertise in: coating and surface technology (PRIREV); coating equipment provider (INTERPANE); air filtration and face masks producer (GVF; industrial textiles coatings (SIOEN) for mattresses, protective clothing, UV coatings for wallpaper and furniture; coatings for high-traffic objects- handles, handrails, etc., and sanitaryware-taps, etc. (PRIREV).
'Surface adhesion and durability via assessing performance against wear (e.g. abrasion, washing, etc.) and degradation in the application environments on a variety of surfaces (e.g. glass, metals & various alloys, copper & steel, marble & stone slabs, ceramics & tiles, textiles & plastics)': **KPI2** High durability of the coatings functionality **KPI5** Proven effectivity and durability in relevant environments **O3** To acquire and apply new knowledge to guarantee the durability and functionality of the coatings**O4** To obtain scientific evidence of impact of the new Nanobloc coatings in relevant environments **O5** To demonstrate the viability of industrialization and the application the new Nanobloc coatings in relevant substrates & products **O8** Validate the developments under relevant conditions & environments	• Some coatings (e.g. POLITO Ag/silica co-sputtered coating) have been **already tested against abrasion and washing and degradation in working conditions** on **several substrates (glass, metal, polymers, ceramics, wood, textiles)** for different **applications** (food processing tools, mobile phones, medical vests, face masks, air filters, biomedical devices,…), giving very encouraging results. They can withstand **temperatures up to 450 °C** without altering their antimicrobial properties, thereby suited for thermal regeneration[9]. • In NanoBloc®, we will perform exhaustive studies to produce **relevant demonstrators and to assess their performance** by testing under standards for each application, as well as testing performance in application environments, with the guidance of the end-users actively participating in NanoBloc®. Moreover, we will develop new testing protocols that will serve as base to contribute to the development of new standards. • In NanoBloc®, thanks to the participation of a **very strong and relevant industrial value chain,** we will be able to produce **prototypes** that will allow us to perform testing and validation work to assess the performance of the coating against wear and degradation in their working environment. • **Demonstrators** will be porous **filter** materials (air filtration units, face masks), **textiles** (protective clothing, mattress covers, aprons, wallpaper), and also **high-traffic solid surfaces** (doorknobs, handles, handrails, sanitaryware-taps, etc.).
'Toxicity of nanocoatings' **KPI4** Non-toxic and Ecotoxic effects of the coatings **O2** To acquire a deeper understanding of the mechanisms of antiviral & biocidal effects	• Some coatings (e.g. POLITO Ag/silica co-sputtered coating) **have been already demonstrated completely environmental- and human health- safe**: Ag nanoparticles are safely embedded in a strong silica matrix. These coatings allow a gradual release of ions without dispersing the nanoparticles in the surrounding environment. Studies on skin penetration have been done, demonstrating no permeation of Ag nanoparticles both in health and wounded skin[10]. • We will perform exhaustive studies to ensure there is no negative impact of NanoBloc® nanocoatings on human health or the environment, and in particular to substantiate with an evidence base of the corresponding **Unique Selling Points of the NanoBloc® coating system** being completely safe for human health and environmentally friendly.

Going beyond the state-of-the-art

State-of-the-Art Baseline

Antibacterial and biocide coatings have been a very active field of research due to their interest for applications such as implants and antifouling. The incorporation of biocidal agents in the coatings is also one of the most used strategies to inhibit the biocontamination of various materials, as the agents can be gradually released into the surrounding environment and minimize or inhibit the colonization of microorganisms on its surface[11].Various additives and biocidal agents have been studied in order to **avoid biocolonization of surfaces**. These include antibiotics, quaternary ammonium ions, inorganic compounds and particles (e.g., Ag, ZnO, TiO_2, nanocarbon,

[9] Chemical, Mechanical, and Antibacterial Properties of Silver Nanocluster–Silica Composite Coatings Obtained by Sputtering, Monica Ferraris et al, Adv Eng Mat 2010, 12, (7).
[10] Antimicrobial functionalization of cotton fabric with silver nanoclusters/silica composite coating via RF co-sputtering Technique. Irfan et al.a, Cellulose (2017) 24:2331–2345
[11] Olmo et al (2020). Antibacterial coatings for improving the performance of biomaterials. Coatings, 10(2), 139.

graphene oxide, SiO_2, carbon nanotubes, etc...)[12]. The use of nanomaterials is also an option, such as silver nanoparticles, zinc oxide, titanium oxide, iron, graphene, copper oxide and metallic nanocomposites[13].**Although significant efforts have been made to develop antibacterial surface coatings, the literature remains scarce for a systematic study on broad-range antiviral coatings**. The scientific literature on biocidal coatings is mostly based on antibacterial mechanisms; **there are only a few pre-pandemic results on viruses**. Copper alloys have shown potential to inactivate Murine Norovirus in less than an hour[14] and can be effective for eliminating human coronavirus 229E[15] in the range of minutes. A few approaches in the literature have explored the possibilities of using coatings with biocidal agents to eliminate virus. In this sense, cold spray has also shown its effectiveness against influenza virus[16] and the potential of Cu, Ag and Zn combined as dopants embedded on sol-gel coatings against enveloped virus[17] have been shown to be effective in reducing the time in which the virus remains active on the surfaces. **POLITO, who is leading NanoBloc®, has developed Ag/Silica nanocoatings by PVD co-sputtering and demonstrated their efficiency (99.99%) to inactivate viruses, including SARS-CoV-2, which is just one of the innovations in our nanocoatings technology platform.**

Products in the market with doubtful claims unacceptable shortcuts & gaps on their development cycle

While there have been no significant previous research efforts focused on coatings to be specifically effective against viruses, the global COVID-19 pandemic has attracted considerable attention towards innovative methods and technologies for suppressing the spread of viruses, and the past 18 months has seen a range of antiviral coatings reach the market. As examples, Table 3 provides an overview of a selection of these products classified according to technology type. The testing of most of these products has been performed against known antibacterial agents with positive results and consequently products have been promoted to market based on a number of **basic standard tests, sometimes including testing against viruses.** Standard tests, while they do give an indicator about the effectivity, are not enough[18]: **more solid evidence is necessary to have reliable coatings, to support the claims and to offer guarantees on a solid base.**

A few of these products have been tested against SARS-CoV-2, according to different protocols; most of them claim antibacterial/antiviral properties. However, while they do provide antiadhesion properties, i.e. microorganisms do not adhere on their surfaces, **they are not killed; and this is a very important limitation.** In other cases, their claims on efficiency against virus, although supported by laboratory reports, **do not clearly show the efficiency when compared to controls**. They acknowledge durability of less than 1-2 years, or simply state that they are durable **without providing any support- or poor justification- of durability claims**.

Table 3- Nanotechnology-based antiviral products

Technology	Brand	Country	Applications
Silver ion technology	Biomaster	UK	Textiles, Plastic, Surfaces, Coatings- e.g. SafeTouch (masterbatch, liquid and powder formulations)
	Microban®/ SilverShield®	USA	Simple spray application
	Italtherm	Italy	Air filtration systems
	Serge Ferrari Group	Italy	Composite membrane to be applied to various substrates
Copper-based technology	HeiQ MetalliX	Switzerland	Vapour deposition process suitable for application on textiles only- air filters, medical gowns, curtains, drapes; clothing, upholstery, mattresses
	Sonovia Technology	Israel	Ultrasound application that impregnates fabrics with Cu and Zn oxides suited for textile finishing
	Bio-BlocXTM	Israel	Face masks
	MedCu	Israel	Face masks
	Pharm2Farm Pro-Larva	UK	Face masks
Other technologies	Maya Sticker	Israel	Adhesive film for application on face masks composed of nano-fibres soaked with the antiviral agent povidone iodine
	ViriMASK	Israel	Face masks based on mechanical multilayer filtration
	AIRDAL®	Germany	Coating based on silica
	Ti-energy® yarn	Italy	Textiles (zinc oxide nano particles in yarn fibers).

Although supported with some laboratory tests, **none of these products on the market have been validated in relevant environments (TRL6)**, such as hospitals, as such studies have so far only been done with copper alloys. **Such shortcuts to introduce products in the market** have been driven in pursuit of short term benefits, and marketing impact & positioning; **this would never be accepted in sectors such as automotive or aeronautics, where the development cycle and careful steps to validate each TRL for the specific application and use are always mandatory.**

[12] Zhang E, Zhao X, Hu J, Wang R, Fu S, Qin G. Bioactive Materials Antibacterial metals and alloys for potential biomedical implants. Bioact Mater 2021;6:2569–612.
[13] CLOUTIER, Maxime; MANTOVANI, Diego; ROSEI, Federico. Antibacterial coatings: challenges, perspectives, and opportunities. Trends in biotec, 2015, 33 (11), p. 637-652.
[14] Warnes SL et al. Inactivation of murine norovirus on a range of copper alloy surfaces is accompanied by loss of capsid integrity. Appl Environ Microbiol. 2015;81(3):1085-1091
[15] Warnes SL, Little ZR, Keevil CW. Human Coronavirus 229E Remains Infectious on Common Touch Surface Materials. mBio. 2015;6(6):e01697-15. 5
[16] Sundberg et al (2015). Effectiveness of nanomaterial copper cold spray surfaces on inactivation of influenza A virus. Journal of Biotechnology & Biomaterials, 5(4), 1
[17] Hodek et al. Protective hybrid coating containing Ag, Cu and Zn cations effective against human immunodeficiency virus and other enveloped viruses. BMC Microbiol. 2016;16
[18] Pietsch, Franziska, et al. "Selection of resistance by antimicrobial coatings in the healthcare setting." Journal of Hospital Infection 106.1 (2020): 115-125.

There is currently a **knowledge gap** in the understanding of the inactivation mechanisms, the optimal application conditions, the effect of the nanocoating in the virus inactivation process, and the effects and damages on materials exposed to these coatings. Three basic mechanisms are responsible for the antibacterial properties of the coatings[19], namely: Anti-Adhesion/Bacteria-Repellence; Antibacterial Agent Release; and, Contact-Killing. Most of the developments are **limited by the early release of the biocidal agents** incorporated in the coatings, which reduces their protection over time. Also, since biocidal agents pass into the environment, their release rate is a key parameter that determines their environmental impact, their potential effects on health, and the rate of onset of antimicrobial resistance. As a result, **the use of biocides and the corresponding concentrations is restricted**.

Understanding how the properties of coatings and biocides determine their release rate is crucial to understanding their durability and enabling the development of effective and safe new products. *Erich et al.*[20] have carried out a study with the aim of establishing this connection through modeling of the release process and some studies show works focused on the design of new coatings capable of **modulating the release rate of biocidal agents by using different technologies**[21,22,23.] Contact-killing mechanisms have been recently proposed and are a field of active research as they have a great potential for **higher durability and lower environmental impact**. The main ones are associated with covalent anchoring of chains on the surface or Cu on the surface. In the first case, mechanisms of action are based on membrane interactions, such as physical lysing or charge disruption, and the most effective compounds for contact-killing coatings have been either cationic compounds (Quaternary Ammonium chains, chitosan, etc.) or enzymes[24]. In the case of Cu, the oxidation of Cu generates Reactive Oxygen Species (ROS) that are **able to kill bacteria even without attacking the membrane**[25]. Although some studies have analyzed this approach, contact killing mechanisms are still not clear and must be differentiated from ion release mechanisms.

NanoBloc®: A ground-breaking nanocoating technology platform

We are advancing the state-of-the-art via our exceptional ground-breaking R&I in the field of innovative virucidal, biocidal and antifungal nanocoatings:

Physical Vapour Deposition: Scalable, versatile and green technology

A key technology that will be used in this project is the industrially scalable, versatile in terms of composition / shape of the surface to be coated and of the composition of the coating and green co-sputtering technology.

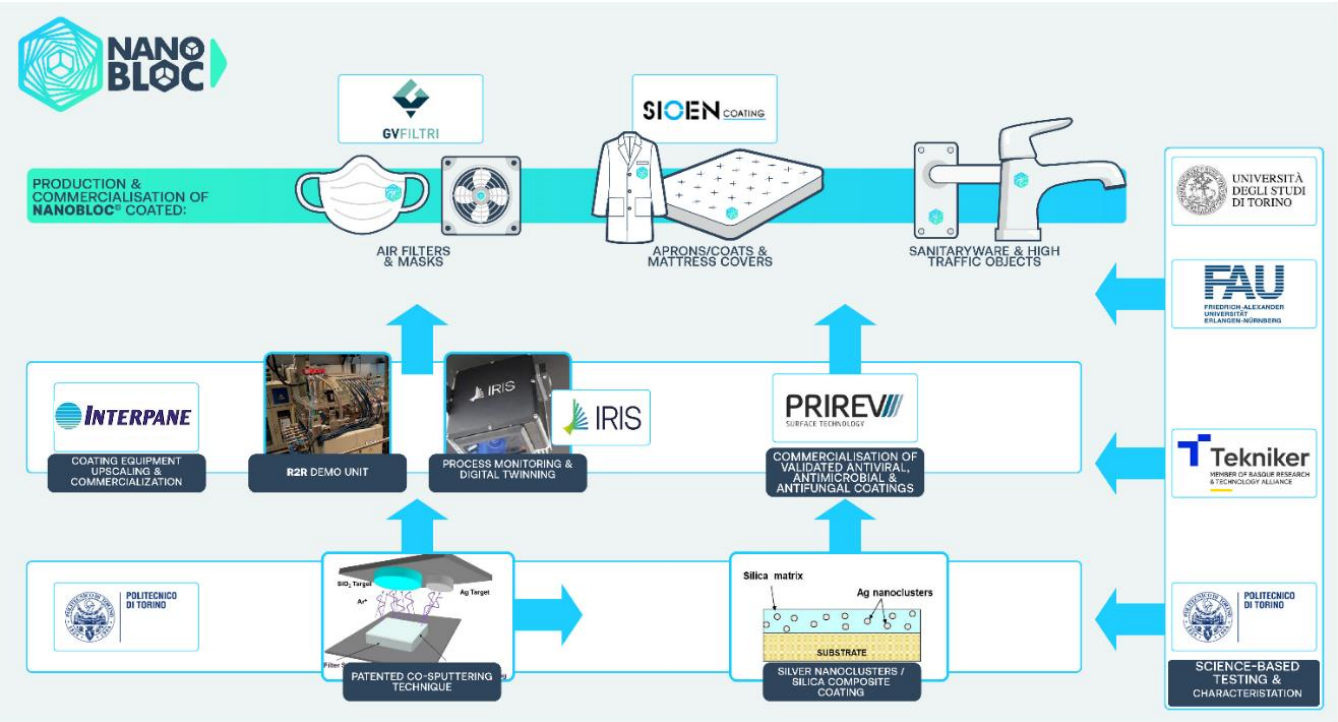


Figure 2- PVD technology and market pathways: example for Ag/silica Co-sputtering

[19] CLOUTIER, Maxime; et al.. Antibacterial coatings: challenges, perspectives, and opportunities. Trends in biotechnology, 2015, vol. 33, no 11, 637-652.
[20] Erich SJF, Baukh V. Modelling biocide release based on coating properties. Prog Org Coatings 2016;90:171–7.
[21] Sikder et al. Development of single-phase silver-doped antibacterial CDHA coatings on Ti6Al4V with sustained release. Surf Coat Tech 2018;342:105–16 Xu Q et al. Antibacterial coatings based on polycaprolactone and polyurethane with prolonged release of ciprofloxacin. Surf Coat Tech 2021;405:126584
[22] Liu et al. Polyelectrolyte multilayer coatings for short/long-term release of antibacterial agents. Surf Coat Tech 2020;393:125696
[23] Zhang et al A sol–gel based silver nanoparticle/polytetrafluorethylene (AgNP/PTFE) coating with enhanced antibacterial and anti-corrosive properties. Appl Surf Sci 2021;535:147675.
[24] Babutan, Iulia et al. "Antimicrobial Polymeric Structures Assembled on Surfaces." Polymers vol. 13,10 1552. 12 May. 2021
[25] VINCENT, M., et al. Contact killing and antimicrobial properties of copper. *Journal of applied microbiology*, 2018, vol. 124, no 5, p. 1032-1046.

POLITO has patented (PCT/IB2018/057639, POLITO) an industrially **scalable, versatile and green co-sputtering technology** that is suitable for deposition on a variety of different solid surfaces (metals, glasses, ceramics, polymers), porous materials, filters, and textiles. The thin coating (less than 200 nm) is composed of Ag nanoclusters well embedded into a SiO_2 matrix. The coating composition can be adapted to the application requirements. It is characterized by a **proven** antimicrobial/virucidal effect and a gradual controlled metallic ion release without the dispersion of nanoclusters into the surrounding environment or human skin. The coating prevents the contamination and proliferation of bacteria, fungi, moulds and it is able to inactive numerous viruses (respiratory syncytial virus and influenza virus A, and **also towards SARS-CoV2**[26] with a study on disposable FFP3 masks) and withstands up to 450°C (suitable for high-temperature cleaning metallic/ceramic/glass substrates; metallic air filters, etc.) retaining its biocidal properties.

- ✓ 99.99% effectivity against SARS-CoV2 and other respiratory viruses, bacteria & fungi.
- ✓ High temperature resistance
- ✓ Nanoclusters not dispersed in the environment
- ✓ Do not penetrate the human skin
- ✓ Successfully sputtered and validated face masks (published results)
- ✓ Suitable for application on a wide variety of substrates
- ✓ Green Technology

TEKNIKER has developed PVD coatings with different technologies (HIPIMS, sputtering, and Cathodic arc) based on metallic nitrides using Cu as biocidal agent, showing that high antibacterial antiviral effects can be obtained for applications such as implants (avoiding toxicological effects)[27] and high traffic objects[28]. **PRIREV** has developed PVD formulations containing Ag with antibacterial properties certified by Hohenstein (PRIREV AB20), that still require adaptation to different substrates to guarantee durability. It is being applied in door handles and is under certification process for food contact to use in cutlery items (to meet hotels and catering demand). The coating is currently in "inox" colour and is necessary to develop films with other colours, requested for decorative purposes such as Black, Gold, Copper and Dark Chocolate. The same properties need to be achieved in substrates commonly used in faucets area, and which are very different in relation to deposition conditions.

Organic/inorganic nanocomposite coatings

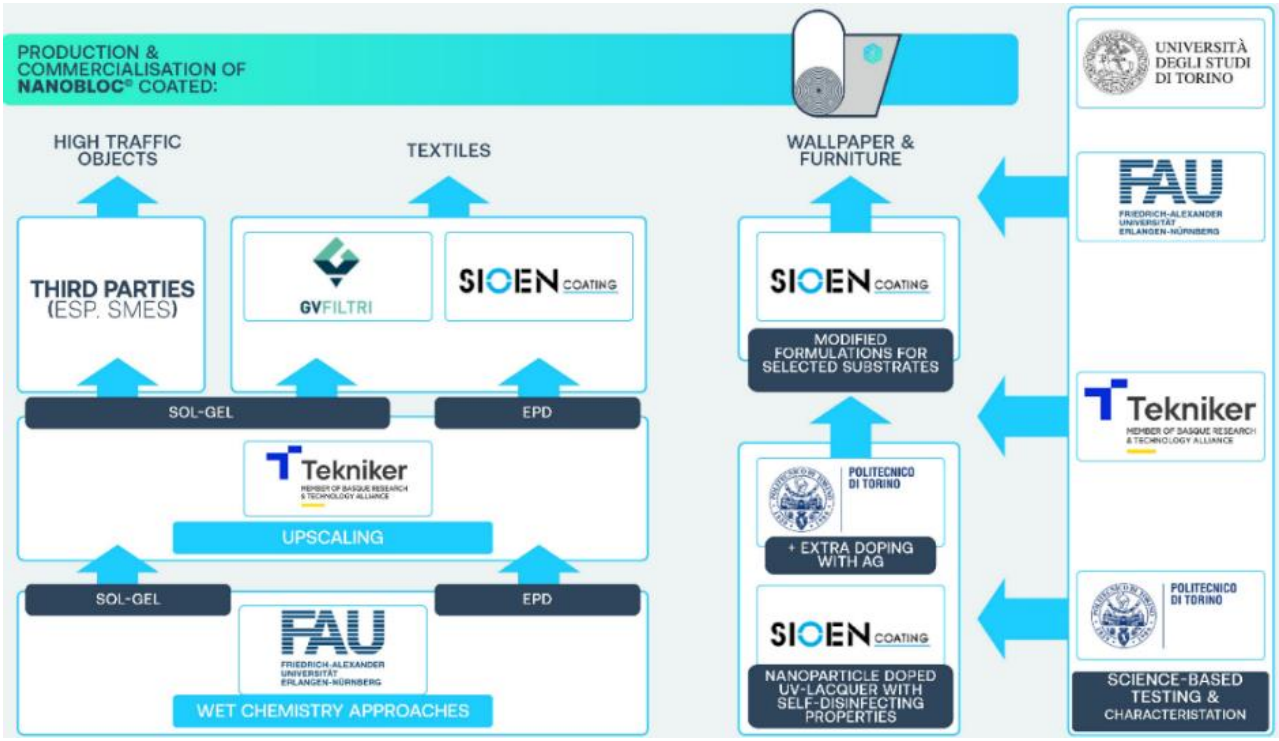


Fig. 3- Technology and market pathways for Wet Chemistry approaches & Nanoparticle doped UV-lacquer with self-disinfecting properties

Sioen Chemicals has more than 20 years of experience in formulating radiation curable coatings. Radiation curable coatings dry (= polymerize) after exposure to UV-light or Electron Beam radiation. This ecofriendly system is **free of volatile organic solvents**, all components stay within the film. Antibacterial products have also been developed

[26] BALAGNA, Cristina, et al. Virucidal effect against coronavirus SARS-CoV-2 of a silver nanocluster/silica composite sputtered coating. Open Ceramics, 2020, vol. 1, p. 100006.
[27] "Desarrollo de una solución superficial antimicrobiana, personalizada y durable que prevenga los procesos infecciosos en implantología dental: IMPLANTIBAC". IDI-20191339
[28] "Materiales multifuncionales de GRAN área basados en la combinación de REcubrimientos y teXturización de superficies: GRAN-REX". IDI-20171116

employing inorganic loads in the polymeric matrix for textile products.[29] Traditionally, additives with biocidal properties can be added as a liquid or solid into the coating system. Conventional products like triclosan, zinc-pyrethion, quaternary ammonium-based products, carbamates, nano-based Ag, all have **disadvantages when it comes to durability in the end product application** as they wash out or lose their efficiency after contact with cleaning chemicals.

Wet-chemistry coatings

FAU has longstanding experience researching on wet-chemistry coatings using technologies such as Electrophoretic deposition (EPD) and Sol-Gel processing to obtain antibacterial properties. As recent examples, they have developed Cu(II)-Chitosan[30] and Ga(III)-Chitosan[31] complex coatings deposited by EPD with high antibacterial activity and mechanical stability. In the case of Sol-gel coatings, they have developed hybrid coatings containing borosilicate glass particles that show good antibacterial properties (with and without Cu) and good corrosion resistance[32]. TEKNIKER has also developed sol-gel coatings that have proven their capability to kill bacteria and virus using Cu as biocide element[33]. The combination of sol-gel coating method and EPD has also been introduced by FAU as a promising strategy to achieve antibacterial coatings on planar and non-symmetric surface geometries[34].

Although the NanoBloc® partners have top level state-of-the art results, further research is necessary in order to **guarantee the desired biocide functionality**, with **adequate durability** for the expected conditions of the applications and with a **non-harmful effect**. In this sense, NanoBloc® will advance the SoA via our **exceptional ground-breaking R&I** in the field of innovative virucidal and biocidal nanocoatings by means of a cocreation approach **that overcomes the main problems of current products in the market while generating solid scientific knowledge on the relation between materials science and microbiological behaviour.**

Thanks to a **powerful cocreation approach** and to **new knowledge that will be generated during the project in relation to the formulation of coatings connected with antiviral, and biocidal and antifungal efficacy of the coatings over time and in operating conditions with a higher durability**, we will select **the most suitable technology from our versatile nanocoatings platform** for each industrial application (based on industrial specification and market requirements) and develop and **up-scale** more effective and durable biocidal coatings (compared to the state of the art) to relevant **demonstrators** that will impact the market in a wide variety of innovative new products and applications for the prevention of airborne infectious disease: porous **filter** materials (air filtration units, face masks), **textiles** (protective clothing, mattress covers, aprons, wallpaper), and also on a variety of **high-traffic solid surfaces** (door knobs, handles, handrails, sanitaryware-taps, etc.), which will be validated in the application environments, including a hospital (San Luigi Hospital that hosts the UNITO laboratory)

As such, as part of a **Cocreation Lab** that will be established and run throughout the project, we will drive forward an application oriented-development that **combines bottom-up with top-down approaches to bridge the gap between science and market needs** in order to develop **new NanoBloc® coatings with scientifically proven superior and reliable properties** in terms of efficiency, durability, and eco-friendliness. A key innovative feature of our project is to implement a 'Quality-by-Design' approach to create a Digital Twin of each process for (i) **increasing the process understanding** thanks to Process Analytical Technology (PAT)-based real-time monitoring for obtaining critical material composition and process data, (ii) **using Machine Learning techniques for predictive model control for process optimization**, which will facilitate the process development and upscaling process, and will also provide the template for a PAT-based process monitoring and control platform for ensuring the standardized processing of the nanocoating during the future (post project) production stage.

This science-base in terms of coatings and the **Artificial Intelligence enhanced understanding of process parameters** (also inspired in the PAT-based Quality-by-Design approach), in addition to materials science properties testing and microbiological behaviour will generate the comprehensive knowledge needed to inform the selection, scaling-up and validation of the most suitable technology for delivering reliable, robust and functional products for the intended applications. **This will ensure that we can make highly credible and scientifically substantiated claims compared to the current State-of-the-Art, whereby many of the current market entrants have been developed to serve a sudden market demand (without sufficiently taking into account the specific needs to each product/application), and tested and validated using simple standards leading to products flooding the market with doubtful claims on functionality.**

[29] https://sioen.com/en/news/technical-textiles-save-lives-mattress-covers
[30] AKHTAR, Muhammad Asim, et al. Electrophoretic Deposition of Copper (II)–Chitosan Complexes for Antibacterial Coatings. Int J Mol Sci, 2020, vol. 21, no 7, 2637.
[31] Akhtar, et al. "Electrophoretic deposition and characterization of functional coatings based on an antibacterial gallium (III)-chitosan complex." *Coatings* 10.5 (2020): 483.
[32] Balestriere, et al. Sol-gel coatings incorporating borosilicate bioactive glass enhance anticorrosive & surface performance of stainless steel implants. *J Electroanal Chem* 876 (2020): 114735.
[33] https://umbrellatch.com/productos/umbrella-zero/. Coating developed by TEKNIKER and transferred to the company: Internal track of background is kept by TEKNIKER.
[34] Ballarre, *Surf. Coat. Technol.* **2020**, 381, 125138

R&I Maturity

In line with the call requirements in relation to Technology Readiness Levels (TRLs), the technologies addressed in the project are currently situated at TRL3 (i.e. experimental proof of concept) and will be scaled to TRL4 (i.e. technology validated in lab), and then through a Living Lab approach in cocreation with the industrial technology upscale partners will be brought TRL5 (i.e. validation in an industrially relevant environment) and then in collaboration with the early adopters to TRL6 (i.e. technology demonstrated in application relevant environment). A key feature of our project is that we are incorporating an AI-driven Quality-by-Design approach to facilitate the process development and upscaling process. Table 4 shows TRL levels of the core components of our NanoBloc® Technology Platform, with current TRL levels, versus that expected at project completion. Further details on how each TRL level is addressed are shown in the methodology section.

Table 4- TRL positioning of the work

Technology	TRL at project start	TRL at project end
Ag/silica co-sputtered coating	TRL4: POLITO has developed the Ag/silica coatings prove to be antiviral & antibacterial and already deposited in face masks at laboratory level.	TRL6 Selected coatings based on their performance on go/non-go decisions will be validated under specific product specifications & shown to be still effective under ageing conditions, & proven performance on relevant environments such as hospital environment & shared areas in working environments (meeting rooms, toilets). Prior to the down selection for final demonstration all the technologies will be validated for up-scaling at relevant environments at TRL5
PRIREV AB20-based coatings	TRL4: Prirev has developed AB20 antibacterial coating in inox colour for stainless steel substrates, however films with other colours and in other substrates need further research	
Cu based PVD coatings	TRL3: TEKNIKER has developed PVD coatings based on nitrides and oxides that are effective against bacteria and virus. Further research needed to develop new coatings compositions & architectures guarantee durability and non-toxicological effects	
Self-disinfecting Nanoparticle doped UV-lacquer	TRL3: SIOEN has developments on antibacterial UV-curing lacquers for textiles applications. Conventional products like triclosan, zinc-pyrethion, quaternary ammonium-based products, carbamates, nano-based Ag, all have durability problems in the end product application as they wash out or lose their efficiency after contact with chemicals.	
Coatings based on SiO_2 & Ag nanoparticles by EDP	TRL3: FAU has designed multi-structured coatings consisting of PEEK/BG layers deposited via EPD and coated with Ag/SiO_2 nanoclusters. Although the slow release of Ag ions resulted in strong antibacterial properties and cytocompatibility, further investigations against various pathogens are required.	
Wet chemical sol-gel process	TRL3: FAU has developed antibacterial sol-gel organic-inorganic hybrid coatings with borosilicate or Cu-doped borosilicate bioactive glass particles. Long-term antimicrobial efficacy of coatings still to be verified.	

1.2 Methodology

Overall methodology.

Cocreation Lab- industry-research codesign innovation approach: A key strength of the NanoBloc® project is that (through the project partners/consortium) we are mobilising a research-industry codesign approach that will be implemented via a Cocreation Lab. Cocreation relates to strategies of open innovation that actively involve the participation of customers in the design and construction of goods and services, in which ideas are produced, valued by organizations and the customers themselves, i.e. cocreation is the participation with customers in order to get value. Our NanoBloc® codesign approach (that can be readily applied to any technology) is underpinned by a pioneering 'cocreation cycle' (figure 4) that will be implemented in the NanoBloc® project in 3 development loops (refer to figure 9 in section 3.1). A key value of this approach is that it is underpinned by research to gain an advanced understanding of the interactions between **process parameters, coating composition and architecture, biocidal properties and mechanisms, in addition to the durability and other properties of the coating**. A major advance will be the encapsulation of this knowledge in self-learning algorithms based on data analytics and machine learning, which will lead to an enhanced understanding and predictive capability over time. Since a digital twin -in comparison to traditional models and simulations that are mainly based in first principles and simple empirical data- is a **data-driven multivariate model, particularly suitable for describing complex systems**, such an approach is of extreme value in the case of processes that involve nano-scale interactions, which are, by their very nature, **too**

- NanoBloc© cocreation process initiates with an application-oriented development philosophy (**products 'ask'**). It informs the selection of the best assessment of the most suitable nanocoating technology in each case.
- Scientific & inter-disciplinary knowledge in relation to Process Parameters/Coatings, Materials Science properties testing, & Mechanical Behaviour enables the most suitable technology and process to be defined (**science 'answers'**).
- The selected process in scaled up, engineered and validated (**engineering 'implements'**).
- The most suitable & adapted technology can deliver reliable, robust & superior functionality products for the intended applications **('technology delivers')** to supply the '**market'.**

complex to be easily described by merely using first-principles-based quantitative descriptions.

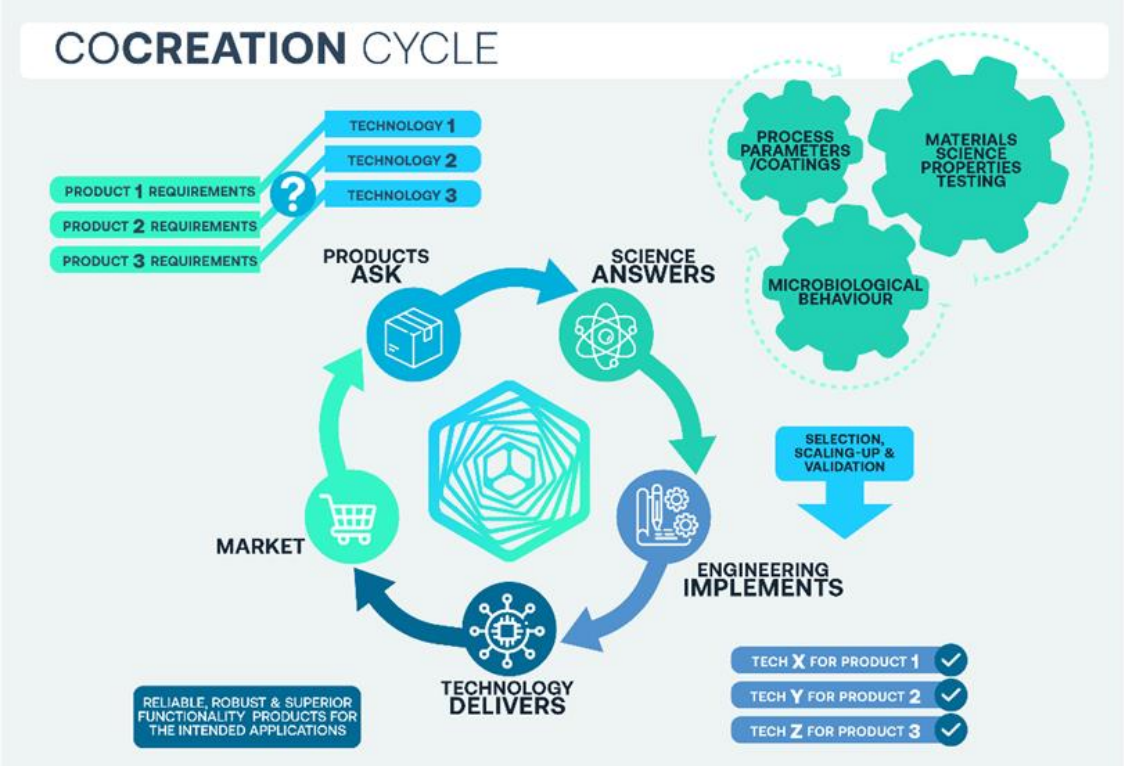


Figure 4- The NanoBloc® cocreation cycle

In terms of the running of our Cocreation Lab, it will be inspired in the 'Living Lab approach', whereby in our codesign and scoping sessions (which will be organised at regular intervals throughout the workplan), in addition to the established theories of innovation, which distinguish two forces- 'technology push' and 'market pull'-, we will also take into account a third force- the **'contextual push'** or the **'people insights push'**. This ensures that we will not only gain the evaluative power (prove/disprove a hypothesis or idea) of our Cocreation Lab, but will also benefit from **generative value (i.e. that we gain access to insights not yet known to the researchers)**. For running our Cocreation Lab (which will cross-cut our entire R&D workplan through Task 4.1), we will follow the best practices of the Living Labs Harmonization Cube[35], which covers the 6 dominant Living Lab perspectives: user involvement, service creation, infrastructure, governance, innovation outcomes, and methods & tools.

Implementation of the NanoBloc® methodology: We will implement the NanoBloc® methodology (i.e. our cocreation cycle) to enable the **selection of the most suitable technology for each industrial application (based on industrial specification and market requirements). It will also facilitate the development and upscaling of more effective and durable biocidal coatings (compared to the state of the art), resulting in relevant demonstrators** that will impact the market in a wide variety of innovative new products and applications for the prevention of airborne infectious disease.

1) Product demands properties

We will produce a comprehensive suite of demonstrators and the performance requirements of each will be defined by the industry end-users in keeping with market requirements. This information will establish the KPIs that need to be verified by our testing and characterization work. Table 5 provides an overview of these **demonstrators** as well as the expected performance requirements of each product as defined by the industry partners. We also indicate the envisaged coating technologies to be applied in each case from among the versatile NanoBloc platform of technologies (and which will be verified during the project cocreation cycle).

Table 5- Demonstrators and performance requirements

Applic.	Product	Performance Requirements
Air filtration	Metallic or glass based filtering devices	▪ Reusable filters made of woven/nonwoven metallic or glass with proper HEPA characteristics and suitable thermal resistance ▪ Capable of withstanding thermal regeneration procedures from 130-350 °C. ▪ Antipathogen properties are superior to market available products, especially temp. resistance (regeneration after heating up to 450 °C). ▪ Effective in killing bacteria, yeasts, moulds, and virus

[35] Mulder, Ingrid & Velthausz, Daan & Kriens, Martijn. (2008). The Living Labs Harmonization Cube: Communicating Living Lab's Essentials.

		▪ No ecotoxic release of nanoparticles into environment
	Face masks	▪ Reusable face masks made of woven/nonwoven textiles ▪ Antipathogen properties are superior to those already existing on the market ▪ Effective in killing bacteria, yeasts, moulds and virus ▪ No release of nanoparticles into environment
Textiles	Protective clothing Mattress covers Aprons Wallpaper	▪ Biocidal according to ISO 21702 (antiviral), ISO 27447 & 22196:2007 (antibacterial) ▪ Maintain biocidal properties for 3-5 years ▪ Temperature/UV stability to hot Xenon testing (black panel temperature of 100°C) ▪ UV stability by means of QUV-A testing for outdoor applications ▪ Biocidal properties are compliant with at least 50 washing cycles at 60°C (protective clothing) or 95°C (medical applications) ▪ Chemical resistance against disinfecting agents such as ethanol, bleach, etc. ▪ Homogeneous biocidal properties at every cm2 over the whole coating width
High traffic solid surfaces	Door knobs Handles Handrails Sanitaryware-taps Window glass Cutlery (buffet/ catering items) Tiles/ Flooring	▪ Effective in killing bacteria, yeasts, moulds and virus ▪ No release of nanoparticles into environment ▪ Resistance to chemicals (cleaning procedures). ▪ Resistance to oxidation (humid environments) ▪ Scratch resistance ▪ Biocidal properties according to ISO 21702 (antiviral) and ISO 27447 (antibacterial) ▪ Biocidal properties for at least 10 years ▪ Temperature stability (hot Xenon testing black panel temperature of 100°C) ▪ Ageing resistance of matrix for at least 10 years ▪ Chemical resistance against solvents, cleaning and disinfecting agents ▪ Homogeneous biocidal properties at the cm2 over the whole surface

2) Preselection of process candidates

The NanoBloc® project will **research and develop innovative virucidal, biocidal and antifungal nanocoatings** that will provide a technology platform that can be codesigned and scaled up according to industry specifications and market requirements. A key value of our project, is that **our technology platform is versatile in that it covers the most promising** (i.e. multiple industrially relevant) **coating technologies.** The cocreation cycle approach will aim to verify the most suitable technology for the different applications. Importantly, our consortium includes the **necessary scientific and technological partners to validate** these technologies in line with **their suitability to deliver product specifications and industry & market requirements,** ensuring a technology and market pathway.

The preselection of process candidates has been performed to cover a wide range of technologies on the basis of the previous experience and good results obtained in past research efforts performed by the NanoBloc® (Table 6).

Table 6- NanoBloc technologies

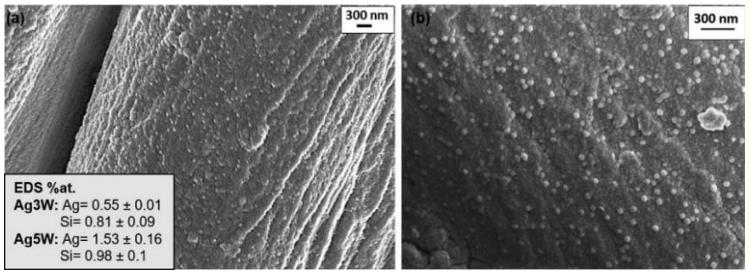
Physical Vapour Deposition	**Organic/inorganic nanocomposite coatings**	**Wet-chemistry coatings**
RF co-sputtered Ag/Silica (POLITO) Sputtering & Cathodic arc of Ag doped nitrides (PRIREV) Cathodic arc, magnetron sputtering & HIPIMs for Cu doped nitrides and oxides (TEKNIKER)	Nanoparticle doped UV lacquers (SIOEN, POLITO) Nanoparticle doped PU & PVC coatings (SIOEN)	Electrophoretic deposition of biobased formulations (FAU) Sol-Gel hybrid coatings (FAU, TEKNIKER)

3) R&D stage (TRL 3-4): understanding the links between process parameters & coating composition/ architecture, microbiological mechanisms & biocidal properties, & coating material properties (durability, etc.)

The properties and functionality of the coatings are determined by their chemical composition, their morphological features, their structure and architecture. In this sense, the physico-chemical characteristics of the coating determine the mechanical and corrosion resistance while the amount and location (at the surface and/or embedded) of biocide elements will determine their antipathogenic functionality in terms of effectivity, depending on the different mechanisms involved, and therefore on the durability and time required to eliminate the pathogens. The physico-chemical features of the coatings are in turn dependent on precise tuning of process parameters and strategy to define architecture (gradient, multilayer, etc.) and to guarantee good adhesion. Some key process parameters for the technologies addressed in NanoBloc® are defined in the following table.

Table 7- Key process parameter for the NanoBloc technologies

Physical Vapour Deposition	Organic/inorganic nanocomposite coatings	Wet-chemistry coatings
Target power & co-evaporation ratios of the different elements in coating composition, frequency & pulse length of the applied power, amount of reactive gasses, temperature, bias voltage, or pre-treatments with ion sources for adhesion, etc.	Type & % of nanofillers, dispersion methods, curing strategy: radiation curable coating- either 100% UV-system (curing by "traditional" UV-lamps and/or Excimer curing) or water-based UV-system (traditional UV or LED-curing), etc.	Electrophoretic deposition: particle size, dielectric constant of liquid, conductivity & stability of suspension, Zeta potential, deposition time, applied voltage, substrate conductivity, … Sol-Gel hybrid coatings: chemical precursors, nanoparticle dispersion, extraction speed (dipping), curing temp, etc.



Sample	TCID50/ml		
	Experiment 1[a]	Experiment 2	
	Ag3W	Ag3W	Ag5W
Coated mask	$(4.5 \pm 4.9) \times 10^3$	$(1.7 \pm 0.6) \times 10^3$	0.0 ± 0.0
Uncoated mask	$(1.6 \pm 0.0) \times 10^4$	$(1.8 \pm 0.9) \times 10^4$	$(1.4 \pm 0.3) \times 10^4$
Virus control	$(2.4 \pm 1.1) \times 10^4$	$(2.1 \pm 0.9) \times 10^4$	$(2.9 \pm 0.7) \times 10^4$

Figure 5. POLITO co-sputtering coatings different composition and morphology leads to different biocide results on SARS-CoV-2

Figure 5 shows POLITO's results on Ag/silica nanocoatings on face masks and how composition (more - Ag5W- or less -Ag3W- silver amount) and morphology lead to different results. Moreover, process parameters influence the growth morphology of the coatings leading to changes on the structure and morphology of the coatings, as illustrated in Fig. 6. While both coatings lead to good biocide properties, the one with densified layers is expected to have better mechanical properties.

In order to understand the relation between process parameters and coating physico-chemical features, **R&D on new coating developments** (which will be performed in **WP1)** will be fed with the results on **testing against different mechanical and ageing conditions (WP3)** as defined by the **applications (WP4)**, to **determine the durability of the coatings, as well as the results on biocide functionality before and after ageing (WP2).** Moreover, an important contribution of NanoBloc® is to understand the mechanisms of antiviral & biocidal effect (WP2) of our coatings in order to improve their efficiency against pathogens, to understand the characteristics that allow them to be wide spectrum antimicrobial coatings and to identify the best fields of application.

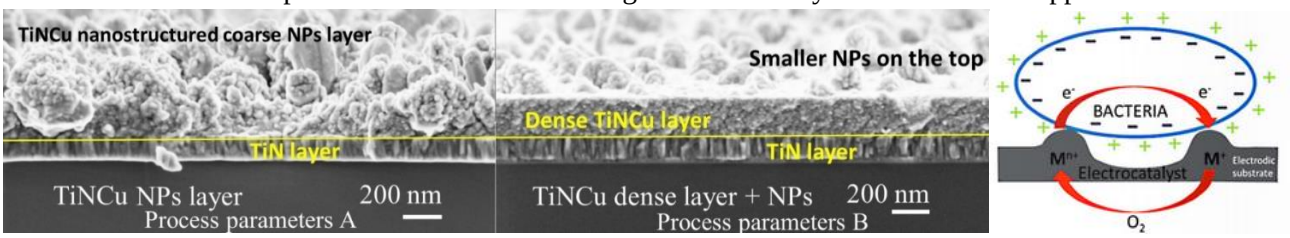


Figure 6. TiNCu biocidal coatings by TEKNIKER. Bilayer structure with different process parameters: Growth morphology is determined by process parameters, leading to nanoparticulated layers(left) or densified layers (center). Scheme (right) of electrocatalytic effect and nanoroughness features of contact killing by generation of ROS.

NanoBloc aims also to focus on the adequate coating durability or the application and on the absence of toxicological effects or environmentally harmful releases. Again, the process parameters and strategies will lead to different morphologies and architectures. As an example, the different surface nanomorphologies of the coatings in Figure 6 lead to different contact angles, and therefore will influence antiadhesion mechanisms, while a control of the coating composition and architecture can provide contact killing by positioning the biocide elements as the Cu at the surface or generating a layered or gradient structure for controlled ion-release. In turn, surface nanomorphology and composition influences electrocatalytic generation of reactive oxygen species (ROS) can kill pathogens by contact[36].

To this end, WP2 will not only feed the development of coatings taking into account their efficiency against pathogens before and after ageing, but also on the different mechanisms acting by employing variations of standard testing to understand touch transfer of bacteria and ion release rates, as well as testing on toxicology and eco-toxicity testing of the formulations to ensure that they are safe. By looping between WP1, WP2, and WP3 a fine tuning of the coatings will be achieved while generating a deep knowledge on the properties and mechanisms acting that will be used to down select the best coatings & processes candidates on go/non-go decisions and selection of strategies for up-scaling and validation for each application.

[36] D. Monaghan et al. Highly biocidal vacuum deposited transparent coatings for touch surfaces to help in the fight of COVID19&HCAI. SVC bulletin.2020

4) Up-scaling & validation stage (TRL5-6): process & properties validation in relevant environments

Thanks to the industry participation in this project, we will mobilise **an industry-driven and application-oriented development that will not only inform and guide the work performed in the R&D phase, it also affords the opportunity to scale up the processes in industry for testing, validating and demonstrating**.

Up-scaling and production of demonstrators

The selected coatings and processes from previous stages will be promoted to up-scaling in order to be validated in industrial environments in terms of technologies (TRL5) and to prepare demonstrators to be tested in relevant environments for the applications (TRL 6). For this purpose, **the selected technologies and coatings will be up-scaled according to the targeted application, and different approaches will be used depending on the substrates and geometries**. Thus, different types of processes will be used to optimize the coatings on the demonstrators. In the case of 3D dimensional objects, batch processes will be used, and processes parameters will be fine-tuned to guarantee the homogeneity of the coatings over the surface of complex shapes such us door handles. It will be necessary to study how positioning and orientation of the samples affect the coating homogeneity, and to design specific sample holders and ancillary tools to guarantee the homogeneity. On the other hand, roll-to-roll processes will be used for 2D applications such as textiles. In this case, controlling web speed together with deposition parameters is key to controlling the homogeneity of PVD and Sol-gel coatings. The methodology of transferring coating is of utmost importance in the case of nanocomposite coatings for high stretch textiles (knitted fabrics and others). In this process, a liquid coating system is applied onto a release paper (temporary carrier) and in the last stage of the process the textile is bonded with an adhesive layer onto the "skin layers".

Table 8 summarizes the types of up-scaling processes in relation to the different technologies addressed in the project and the partners in NanoBloc.

Table 8- NanoBloc Upscaling processes

Batch processing: 3D applications	Roll-to roll processing: 2D applications
-PVD sputtering/cathodic arc/HIPIMS (PRIREV-TEKNIKER) - Sol-Gel (TEKNIKER) - EDP (FAU-TEKNIKER)	- PVD Sputtering (INTERPANE) - Sol-Gel (TEKNIKER) - Nanocomposite coatings & lacquers (SIOEN)

In order to speed up the adaptation of process parameters to industrial environment up-scaling and to guarantee the quality of coatings a data-based strategy using digital twins, in-line process monitoring and machine learning will be used, on the basis of the data obtained from the earlier stages of the project.

Digital twin approach: In-line process monitoring for closed loop control and quality standardisation

A key innovative feature of our project is to implement a 'Quality-by-Design' approach to create a Digital Twin of each process for increasing the process understanding thanks to Process Analytical Technology (PAT)-based real-time monitoring for obtaining critical material composition and process data. Machine Learning techniques will also be used for predictive model control for process optimization, which will facilitate the process development and upscaling process and will provide the template for a PAT-based process monitoring and control platform for ensuring the standardized processing of nanocoatings during the future (post project) production stage.

Based on advances in manufacturing and processing technologies, a key innovation is planned under the Digital Twin/Industry 4.0 paradigm. A digital twin is a digital replica -mainly built on the basis of multivariate empirical data- of an artefact, process or service that is so accurate that it can be used as basis for taking decisions[37]. The concept of the digital twin is one of the big game-changers in manufacturing allowing companies to significantly increase their global competitiveness. It further securitizes IP by adding a control layer that relies on decentralized **decision-making, to support (or even replace) human intervention**. Our intention is to implement Digital Twin technology for ensuring that the scaled-up processes are optimally controlled based on predictive models or protocols that will be developed for the target end products.

The 'nano' tends to 'live' at a holistic level that cannot be accessed directly with spectroscopy (except for very specific cases). However, this holistic level in turn influences higher levels, which are accessible through spectroscopy and other 'extensive' techniques. In order to develop a digital twin for macroscopic levels (i.e., for production purposes), it would be very valuable to have access to microscopic information, for which nanoscale characterization techniques provide data source on different physics and insights on nanoscale features.

A Process Analytical Technology (PAT) enabled control platform (based on the IRIS Scadalytics software platform) will incorporate Artificial Intelligence and Machine Learning modular engines for keeping the process under optical conditions. It holds significant potential for further improving process resource and energy efficiency, and for ensuring that a standardised quality product is obtained based on the production protocols.

[37] https://www.change2twin.eu/digital-twin/

The concept is to design, analyse, and control manufacturing processes through the measurement of identified **Critical Control Parameters** that govern product variability, as well as real-time in-line monitoring of analytical quality attributes of nanocoating being processed. This concept is enabled by installing PAT tools, particularly infrared spectroscopy, in the processing equipment. The philosophy behind the PAT initiative is the **Quality-by-Design principle**, whereby the design of the process relies on a sound and scientific understanding of the relationship between **Critical Quality Attributes** of the product itself and the control parameters (See Table 7). The quality attributes make the product safe and efficacious, and the control parameters influence product quality inside the acceptable process limits. In-line monitoring and control of these parameters can better assure product quality during manufacturing. The quality attributes of a product are directly determined by the control parameters, and as such, the quality of products can be incorporated by process design and not by postproduction quality testing. Furthermore, a **data fusion approach**—i.e., combining spectroscopy with traditional parameters and relating them to quality parameters of the production—will provide a more robust mathematical model automatic control of processes, improving yields, reducing time, waste and energy consumption, and for predictive maintenance. In short, based on real-time data provided by the PAT system in relation to the quality attributes of the nanocoating material, the control parameters governing product variability are modified in real-time based on the models to ensure optimal product quality with efficient process conditions. Data from the PAT analysers will be processed with chemometric (Machine Learning) models in order to extract usable information from the spectral data, and it will be fused with the data from the traditional sensors and processed with advanced models for closed-loop process control. It provides information for improved decision making, sending alerts to operators, as well as using data historian about the process for improved process understanding, optimisation, predictions, & anomaly detection.

Validation of demonstrators in relevant environments

Once the up-scaling stage is optimized, the coatings will be deposited on demonstrators for the different applications. These demonstrators will be validated against the specific standards of each application but additionally, the NanoBloc solutions will be tested in relevant environments (TRL6) to show their effectiveness and durability over 1 year trial campaigns. This will be done by placing coated elements (such as doorknobs) in specific places (toilets, meeting rooms, etc) in a hospital (San Luigo Hospital that hosts the UNITO laboratory) and working environments- toilets, meeting rooms, etc. (TEKNIKER, POLITO). The elements and samples will be periodically analysed by extracting and/or swabbing both coated samples and non-treated samples in equivalent places to analyse if there are significant statistical differences on the pathogens found on treated and non-treated surfaces. Cleaning procedures will be kept as usual in the chosen environments in order to study the durability under normal operations, and in this sense evidence of degradation due to oxidation, cleaning procedures or normal use will be obtained.

How the methodology will enable delivery of the project's objectives

User-centric development and validation can play an important role in speeding up effectively the innovation process through addressing the actual user needs (European Commission, 2009). Whereas users are typically seen as a subject in traditional research, they are seen as **co-creators of innovation in a Living Lab approach**. Our approach ensures that **we bring the industry partners and target end-users into the creative process early in order to discover new and emerging requirements, expectations, behaviours and user patterns**. In this way **we will bridge the innovation gap between technology development and the uptake of new products and services** involving all relevant players of the value network through engagement between the innovation creators (from our RTOs and coating industry), the innovation adopters (end-user industries, including hospitals), as well as citizens (consumers), and enabling stakeholders (standards bodies and policy makers). We will also allow for the early assessment of the socio-economic implications of the outputs by performing impact assessment work. In conclusion, the **advantages of our methodology** are as follows: More reliable coatings/products with better properties; Potential to define new, future-oriented, 'fit-for-purpose' standards for validation; Potential to easily adapt to new demands once key R&D testing are defined, and processing and fundamental knowledge of coatings is developed and shared in Open Science practices and databases; Better social engagement: customer will know that products are based on strong knowledge basis and methodology through dissemination actions to wide public.

From a project management perspective, we will implement the most appropriate methodology in view of our Cocreation approach to the implementation of our project. As it is appropriate for our research-industry codesign/cocreation approach, we will adopt a **hybrid-approach to our project management methodology**, whereby we will combine both the 'predictive' and 'adaptive' project management approaches. In the **predictive approach** (also referred to as 'waterfall'), the **project scope, time and cost are defined in the early phases of the life cycle (and this is the case for our project), and then any future changes to the scope are dynamically managed**. While this approach is effective for managing the carefully planned and budgeted project plan that has been drawn up and defined based on a strong base of experience of the project consortium in designing and executing large-scale multidisciplinary, multi-actor and multifaceted collaborative R&D projects, in situations where projects are dynamic in nature, involve some potentially unknown user requirements, potentially unknown cause and effect interdependencies, and complexity, it can present difficulties in responding quickly. To this end,

when we are presented with these situations, **we will blend in the adaptive (agile) approach**[38], whereby the detailed scope is defined and approved before the start of each iteration. This approach is very fitting for managing the implementation of the 3 development and innovation cycles that will be executing during our workplan. **Our hybrid approach will ensure that for those elements of the project that are well known, we can follow a robust predictive life cycle (waterfall), however (and this is important for the dynamic Cocreation Lab approach), for those elements that are still evolving, we have the agility to follow an adaptive life cycle**. Such a hybrid approach will ensure we can achieve the best outcome.

How the methodology complies with the 'do no significant harm' principle

In compliance with the 'do no significant harm' (DNSH) principle, Table 9 outlines how our project activities contribute substantively to the 6 environmental objectives listed in the Regulation (EU) 2020/852 of the European Parliament and of the Council of 18 June 2020 on the establishment of a framework to facilitate sustainable investment, and amending Regulation (EU) 2019/2088 (Taxonomy Regulation). Thus, we confirm that our action **does not significantly harm any of the 6 environmental objectives and contributes to reach them**. However, we will continually monitor this to ensure full compliance, and in particular this will be assessed in Task 2.3 Toxicity & ecotoxicity. We will also leverage the expertise of regulatory partner UNE.

Table 9- Compliance with the 'do no significant harm' principle

No.	Enviromental objs.	How NanoBloc® contributes substantively?
1	Climate change mitigation	Longer life time of products means a reduction of the processes and materials used having an impact in the reduction of emissions due to manufacturing processes.
2	Climate change adaptation	As many of the root causes of climate change also increase the risk of pandemics, the NanoBloc® innovation will help Europe build resilience in relation to these risks.
3	Sustainable use and protection of water and marine resources	PVD technology is a green & eco-friendly method, based on inert gases, and in the absence of solvents, wastewater or potentially toxic precursors with health and environmental risk. Nanocomposite and wet chemistry technologies addressed are solvent free solutions. Ecotoxicity testing to ensure no harm caused to marine species (see point 6 below).
4	Transition to a circular economy	Waste prevention is key for circular economy. One of the main aims of NanoBloc® is to deliver durable coatings. This will extend the operative life and of products such as face masks and air filters reducing waste of disposable filters and textiles. Ag/silica coating that can be easily regenerated by thermal treatments, will make some products reusable (e.g. air filters), decreasing the quantity of waste to be processed as special waste.
5	Pollution prevention and control	PVD technology is an extremely green technology. Anti-bio-pollutant properties have been proven effective in preventing contamination and proliferation of biopollutants such as bacteria, fungi, moulds & viruses. Radiation-curable coatings dry (i.e. polymerize) after exposure to e.g. UV-light or Electron Beam radiation. As all components stay within the film it is free of volatile organic solvents.
6	Protection and restoration of biodiversity and ecosystems	Gradual controlled release of ions without dispersing the nanoparticles in the surrounding environment is on the targets in the project. Ecotoxicity testing for marine species will be performed and any ion release will be strictly analysed for toxicological effects & ecotoxity to reduce potential harmful ions & reach a balance between the biocide effect against pathogens, while ensuring that no harm is caused in terms of toxicity and ecotoxicity.

Challenges and how they will be overcome

The objectives and methodology proposed in NanoBloc are ambitious, in that a methodology that is sufficiently versatile for different technologies, coatings and applications is addressed. While such an approach has the advantage of covering a diversity of technological solutions, minimizing risks and delivering high value for the money invested, it does have a number of inherent challenges to be overcome, as summarized in Table 10.

Table 10- Challenges associated with our chosen methodology and how they will be overcome

Challenges	NanoBloc approach
To deliver effective and durable coatings for a wide range of different applications and substrates	Having different applications and substrates implies that there are different requirements. While the pre-selected technologies have shown great potential, not all of them may be finally suitable for the applications in a reasonable time. As such, NanoBloc has pre-selected the most promising ones, and the proposed R&D stage of the methodology looping characterization against specific needs (which will be facilitated by the Quality-by-Design approach) will allow to down select the most suitable technology/coating for each application.
To handle the	In order to be able to work on different coatings/technologies and to obtain the expected

[38] Böhmer, et al (2015) Open Innovation Ecosystem - Makerspaces within an Agile Innovation Process. ISPIM Innov. Summit, Int Soc Innov Manag, 1-11

development of a wide range of coatings/technologies in project lifetime	results within the lifetime of the project, leading scientific and technological organizations will work on the different technologies, with wide experience and building on previous good results on each specific technology (e.g. POLITO: Co-sputtering, TEKNIKER: HIPIMS, FAU: sol-gel and EDP, SIOEN: radiation curing, etc.).
To characterize a high number of coatings during development.	Having different technologies and types of coatings could lead to bottlenecks in the characterizations to feedback the developments. In this sense, all the partners have basic characterization tools (such as adhesion) to pre-screen their coatings prior to being sent for more complex testing with lower availability. Additionally, laboratory characterization techniques such as SEM, XRD, etc. are available at different partners (POLITO, TEKNIKER, FAU…), and will be used to cover the physico-chemical characterizations loads of all partners. The characterization loops will have a hierarchical structure based on the complexity and availability of testing rigs. Pre-screening and down selection by go/non-go decisions will be used to only promote the best performing coatings in wider available simpler tests for further testing. Thus, antiviral effectivity and toxicology (UNITO) will be tested on good antibacterial pre-screened samples (done in parallel by TEKNIKER, POLITO and FAU for different coatings).
To up-scale of a wide range of coatings/technologies in the life time of the project	The characterization loops with coating development will allow to down select the most appropriated formulations to be up-scaled. On the other hand, as in the case of the coatings development stage, the same is applicable to the up-scaling stage for which there are existing pilots and industrial facilities for the different technologies among the partners in NanoBloc, so each technology can be up-scaled in parallel by different partners (INTERPANE- sputtering roll-to roll, PRIREV- batch PVD processing, SIOEN- nanocomposite coatings, TEKNIKER- sol-gel coatings). The up-scaling process will be accelerated by the use of data from earlier stages of development in the project and machine learning strategies (in a QbD approach), especially as simulation tools capable of producing reliable predictions about the process behaviour in larger scales. However, as no realistic data from a scaled-up process will be available in the earlier stages, the models for such scaling-up simulations should also comprehend first-principle-based algorithms robust enough to handle any scale-dependent behaviour.
To validate and demonstrate the effectivity and durability of the coatings for different applications on relevant environments	This is one of the main challenges addressed by NanoBloc is validation in relevant environments. This is one of the main advances regarding the state-of-the-art, as current products in the market have not been correctly validated. The cocreation approach is one of our key tools to address this challenge, as developments will be driven from application requirements to establish the relevant testing, and the results will be continuously monitored using hybrid management to address the go/no-go decisions on each coating/technology/application, enabling solid results on the final validation stage, where trials will be done to demonstrate the performance of the coatings on relevant working environments.

Feed in from national or international research and innovation activities

Table 11 summarizes some of the **key EU & national/regional research & innovation activities that are feeding knowledge and results (background) into the NanoBloc® project**. In addition, we will also **promote links between our project and the other projects funded under this topic** (and other relevant Horizon Europe topics), in order to establish a common approach with them in relation to defining certain data sets (to guarantee that they are interoperable), to identify the synergies for **joint dissemination actions (workshops, events, etc.)** and mutually leveraging our communication channels **towards the general public and common target groups**. These links will be managed throughout the workplan leaded by POLITO, supported by all the partners.

Table 11- Relevant projects and the link with NanoBloc®

Brief outline of the results that are feeding in will be used in NanoBloc®
BIOKILLER – antiBIOpollutant coating for reusabLe filter", Manunet III Transnational Call (2018) Deposition and characterization of Ag/SiO2 antipathogen nanostructured coatings via co-sputtering for air filters and metalcutting filters (GVF coordinator; POLITO partner; TEKNIKER and UNITO R&D subcontractors)
FLEXPOL: Antimicrobial FLEXible POLymers for its use in hospital environments (H2020-721062) TEKNIKER was a partner in this project in which they have generated knowledge in roll-to-roll production systems for cost-efficient manufacturing of extensive antimicrobial adhesive films.
NABLA: Nanostructured antibacterial layers, Piedmont Region Call Study and characterization of Ag/SiO2 antibacterial nanostructured coating deposited several substrates for aerospace inflatable modulus, biomedical implants and mobile phone (POLITO, coordinator)
NASLA: Nanostructured antiseptical layers, EU 7th FP project Study and characterization of Ag/SiO2 antiseptic nanostructured coating deposited via co-sputtering on several substrates as biomedical implants, cheese container food

handling equipment and textiles used for personnel protective systems (POLITO, coordinator)
POLYBIOSKIN: High performance functional bio-based polymers for skin-contract products in the biomedical, cosmetic and sanitary industries. H2020-BBI- 745839 Among its objectives was development & scale up of nano-structured highly skin-compatible textiles for wound dressings. IRIS (who also coordinated) will link in knowledge in monitoring of distribution & quantification of nanoparticles applied on new textile films using Hyperspectral Imaging.
MARTe: Antibacterial coating for textiles, Piedmont Region Call: Development and characterization of Ag/SiO2 antibacterial coatings deposited via co-sputtering on cotton and high-performance technical textile materials (Kevlar and Vectran) for aerospace and civil applications. (POLITO, partner)
EcoPUR, GreenWin (Belgium): Research into PU coatings for synthetic textile (specifically for hospital mattresses) in accordance with future environmental and health legislation and to anticipate future production standards. Project aimed to replace organic substances with more sustainable (nano) alternatives.

Interdisciplinarity

A key feature of our project is the mobilisation technology, scientific, engineering, product and market expertise from interdisciplinary fields, along with the cocreation of research and industry partners. Thus, in order to reach NanoBloc objectives it is necessary to bring together different disciplines such as coatings, thin film and nanotechnologies for new coatings development; microbiology for biocide mechanisms and effectivity and toxicological testing; materials science for physico-chemical and durability characterization; engineering, AI & digitalization to ensure the process up-scaling; and industrial partners and end-users to establish and validate the requirements for the applications. Importantly, NanoBloc® also pools together complementary expertise and competencies of partners representing industrial and academic/research excellence in their respective fields.

Interdisciplinary R&D, while it improves creative performance and the potential for breakthrough innovations to be generated, **is complex to manage**. In order to promote the seamless integration of these disciplines during the project, the organization structure will include an **Interdisciplinary Strategic Committee**, involving senior researchers from each of the different expert teams. This Committee will be responsible for the definition of the internal working strategies of the **Thematic Working Groups** from each WP, and for the consolidation of the interdisciplinary R&D programme with the aim of by **minimising redundancy, balancing diversity and complementarity, and capitalizing on synergies in information, knowledge, cultures and techniques.** This Committee- which will be chaired by a **Scientific Manager**, Dr. Borja Coto, an experienced researcher and scientific coordinator from TEKNIKER and expert in Multifunctional Surfaces, industrial processes and scale up- will overcome communication barriers, ensure the seamless transfer of strategic information across the disciplinary teams, improve the risk management and Scientific and Technical decision-making process. Moreover, the tools and methods that we will employ in order to enable the fluid convergence and intersection of these disciplines towards the successful accomplishment of the project objectives are as follows: codesigned work plan (all teams and disciplines have been involved in defining the work plan, and in identifying the necessary interactions required among the various disciplinary teams); Cocreation Methodology (as discussed), digital collaboration and knowledge management tools; holistic plenary meetings (every 6 months); Thematic Working Group sessions (at least monthly); informal breakout meetings and sessions (as often as needed)- with an agile minutes template to be completed for every meeting and session); Internal knowledge exchange, demonstration and training workshops.

Integration of social sciences and humanities: Not applicable for this call topic.

Gender dimension

Gendered Innovations harness the creative power of sex, gender, and intersectional analysis for innovation and discovery, and considering these approaches may can add valuable dimensions to research, in that they may take research in new directions (which would not have unveiled themselves had we not have considered gender dimension). As such, we will include a 'gendered innovation' approach in our project, and through Task 7.4, we will employ methods of sex, gender, and intersectional analysis as a possible source for new knowledge creation. We will produce a publicly available **Gender dimension Report (D7.5)** that will describe how gender dimension has been integrated into the project across all its stages and what the main results and outcomes from a gender perspective have been. Key areas where gender dimension has been integrated into the content of our project are:

- Sufficient (at least 40% representation) participation of female stakeholders, citizens and end-users in the codesign and cocreation toward facilitating innovative solutions; this will ensure the different needs of women and men groups is reflected in the cocreation process for developing the NanoBloc® technologies.
- Perform a gender analysis to consider if there are any differences in how e.g. NanoBloc® coated masks or textile clothing may perform differently through use and different behaviour of women versus and men.
- D&E (Dissemination and Exploitation) outputs will ensure gender inclusive language and images, as well as specific attention to women's and men's (potential) different characteristics, interests, roles, etc., revealed by the project results and which are or should be taken into account in its applications. Gender stakeholders in the

field, as well as women from intersectional groups related to the project application sectors will be considered as participants and targets of the D&E activities.

Open Science Practices

Open science practices (i.e. an approach based on open cooperative work and systematic sharing of knowledge and tools as early and widely as possible in the process) will be implemented as an integral part of our proposed methodology and for increasing our chances of delivering our objectives, as follows:

- **Early and open sharing of research:** Publications will be made in **open access journals** and we will ensure early sharing of publications (pre-prints) to satisfy open access requirements and to enable a faster way of sharing knowledge via deposit in repositories. Participation in open peer-review will also be encouraged.
- **Open access to research data and other related research outputs** will be promoted following the principle 'as open as possible, as closed as necessary' and operated in parallel with a robust strategy for ensuring commercial exploitation, protection of personal data or confidentiality/security are not jeopardized.
- **Research output management:** use of lab books and effective (and digitized where possible) documentation procedures and effective packaging of research results.
- **Measures to ensure reproducibility of research outputs:** Mandatory technical standards will be adhered to in order to ensure that scientific information, publications, data and other outputs from our project, as well as the metadata about them, is available for re-use in the long term. This will involve the use of persistent and unique identifiers, of certified repositories that are compliant to the standards of the **European Open Science Cloud and in** compliance with the FAIR principles for the management of research data produced by projects.
- **Involving all relevant knowledge actors including citizens, civil society and end users in the cocreation of content, as well as Citizen Science:** we will organize a series of webinars/focus groups seminars targeted at Civil Society Organisations (consumer-oriented) and to a wider audience with open sessions for addressing questions and concerns of the citizens that can be taken into account in the cocreation circle.

Data Management and management of other research outputs

Data Management is an integral part of the research process for our project. Table 12 provides an overview of how data/ research outputs will be managed in line with the FAIR principles (Findable, Accessible, Interoperable, Reusable). A full **Data Management Plan (DMP)** will be ready by M6 as D7.2 (it will be updated as project implementation progresses and when significant changes occur). It will specify what data is generated by the project (purpose of the data collection/generation and its relation to the objectives of the project), whether and how it will be exploited or made accessible for verification and re-use, and how it will be curated and preserved to ensure our data/research outputs are findable, accessible, interoperable and reusable. In line with FAIR principles, our DMP will address research outputs other than data & will consider aspects related to the allocation of resources, data security & ethical aspects.

Table 12- Summary of data management and management of other research outputs

Types of data/research outputs and their estimated size In our work plan in Task 4.3, we will be acquiring key data in relation to process parameters (for process monitoring purposes), and we will also be implementing an AI, applying machine learning techniques for model predictive control. In this sense, the management of process parameters data, which is a highly valuable project output, will be addressed. In addition, we will also gather valuable data from the experiments in WP2 and WP3, and the testing protocols themselves. While we may not be able to disclose sensitive process data, we can show ordered data on the results of testing e.g. experimental images, text, numerical, etc.
Findability of data/research outputs We will ensure we make the data readily findable. Using different types of persistent and unique identifiers (e.g. digital object identifiers) and trusted repositories.
Accessibility of data/research outputs We will ensure that the data is deposited in a trusted repository and we make the data openly accessible, as early as possible in the project and in keeping with robust IPR management (data linked to IP will be available after IP protection mechanisms are in place: the aim is making research data available as soon as possible). Provisions will be made for providing controlled access to restricted data for verification purposes, and we will also make provisions for providing access to the EC if requested by the EC in response to states of emergency.
Interoperability of data/research outputs We will use open, international and public available standards, e.g. the standards developed by EU and international standardization organizations like CEN and ISO, such as ISO 10303 (for CAD, CAE/Simulations, PLM and sensor data (IoT) integrations, in addition to data storage for AI/ML applications) that: increase interoperability of information technology, addressed by data exchange & sharing solutions; enable common enterprise-wide views of information, addressed by data integration solutions; address obsolescence of information technology by data archiving solutions; promote freedom from vendor lock-in, addressed by open data solution. **The participation of UNE (representing CEN and CENELEC) in our consortium is a significant asset.** Moreover, other proposals approved in the topic (and other relevant Horizon Europe topics) will be contacted to establish common definitions on data sets and guarantee their interoperability.
Reusability of data/research outputs Where possible, we will make the data produced in the project usable by third

parties including after the end of the project. We will provide the documentation needed to validate data analysis and facilitate data re-use (e.g. readme files with information on methodology, codebooks, data cleaning, analyses, variable definitions, units of measurement, etc.). Where possible, data will be made freely available in the public domain to permit the widest re-use possible. The aim will be for data to be licensed using standard reuse licenses *(e.g. Creative Commons, Open Data Commons)*. The provenance of the data will be thoroughly documented using the appropriate standards, and all relevant data quality assurance processes will be described.
Curation and storage/preservation costs Research data/output management will be managed through Task 7.2 (Data Management and FAIR compliance)- these costs are for the most part personnel costs for the experts involved. A core team with prior expertise in Data Management (and data quality assurance) will be mobilised under the leadership of Dr. David Nettleton from IRIS (a senior Data Scientist) as the Data Manager for the project. A reasonable budget has been planned under 'Other Costs, Works and Services' for making data FAIR and for long term preservation.
Data Security The data will be safely stored in trusted repositories for long term preservation and curation. Provisions will be made for data recovery as well as secure storage and transfer of sensitive data.
Regulatory and Ethical aspects e.g. compliance with GDPR (General Data Protection Regulation) for any personal data gathered during the project.

2. Impact

2.1 Project's pathways towards impact

Unique contribution of project results to (1) the outcomes specified in this topic, and (2) the wider impacts, in the longer term, specified in the destinations in the work programme

1. Outcomes and impacts of our project

Table 13- How NanoBloc® contributes to the outcomes specified in HORIZON-CL4-2021-RESILIENCE-01-20

Expected outcomes:	NanoBloc® contribution
Minimise the risk of spread of infections from harmful pathogens arising from everyday human activities	- All the coatings proposed in the NanoBloc® platform of Physical Vapour Deposition (PVD), Organic/inorganic nanocomposite coating and Wet-chemistry coating technologies have already been demonstrated effective against several **bacteria, fungi and viruses.** In particular, POLITO Ag/silica co-sputtered coating has been tested and is effective on a **wide range of bacteria and fungi & viruses including respiratory syncytial virus, influenza virus A**[39] **and SARS-CoV2**[40], and PRIREV AB20 coating was tested against adenovirus and proved to be antiviral. - When applied on everyday life, high traffic objects and air filtration systems, they will contribute to **minimising the risk of spread of infections from harmful pathogens arising from everyday human activities** thanks to a scientifically proven ability to deliver long-term protection and reduction of risks of infection after being validated in critical environments, such as hospital and healthcare facilities, workplaces, …
Create a healthier living and working environment and offer holistic solutions to people with a diminished immune system	- We will produce **relevant demonstrators and assess their performance** in living and working **environment, specifically focused** on **people with a diminished immune system, such as in hospitals**, but targeting future applications on elderly people care residences, kinder gardens, schools, public transportation,… - We are offering holistic solutions as demonstrators will cover all porous filter materials (**air filtration units, face masks**), textiles (**protective clothing, mattress covers, aprons, wallpaper, furniture, flooring**), and also high-traffic solid surfaces (**doorknobs, handles, handrails, sanitaryware-taps, etc**.).
Improve citizen health and enhance the EU's reputation as a public health best practice Region	- Nanobloc methodology based on cocreation approach that combines functionality, durability in relevant conditions and toxicological/ecotoxicological effects has the potential to develop new to protocols to be promoted as new standards, something that is clearly recognized as a need[41]. Thus, direct results from NanoBloc can provide a wider impact on the future adoption of self-disinfecting surfaces. The knowledge generated during the project will contribute to **defining new and better standards for validation** and **will position Europe as a science-based leader on the world stage in the field of assessing performance of antimicrobial and antiviral**

[39] Nanostructured composite coating endowed with antiviral activity against human respiratory viruses deposited on fibre-based air filters.Balagna et al, Surf Coat Tech 409 (2021)
[40] Virucidal effect against coronavirus SARS-CoV-2 of a silver nanocluster/ silica composite sputtered coating. Cristina Balagna et al. Open Ceramics 1 (2020) 100006
[41] Pietsch, Franziska, et al. "Selection of resistance by antimicrobial coatings in the healthcare setting." Journal of Hospital Infection 106.1 (2020): 115-125.

	nanocoatings and will guide consumers, manufacturers, researchers and policy makers. - Nanobloc coatings developed during the project will serve as a scalable, versatile innovation enabling platform for the development of Antimicrobial, Antiviral, and Antifungal nanocoated products in the future, whereby the NanoBloc® registered trademark will **become synonymous with trusted levels of high efficiency, durability, safe and environmentally friendly coatings.**
Enhance economic benefits through reduction of lost hours of work through illness	The NanoBloc® coatings will provide antimicrobial/antiviral face masks, air filtration, textiles and high traffic objects that will be applicable in workplaces to prevent the spread of illness amongst workers, thus **enhancing economic benefits through reduction of lost hours of work through illness.** Additionally, the potential to reduce illness due to nosocomial infections is huge as quantified in section c below.
Boost research, development and innovation in the EU	**-All the coatings proposed in the NanoBloc® platform are will have been cocreated by all-European research, design, production and (future) commercialization of innovative products**, ensured by a **very strong and relevant industrial value chain, including:** Coating equipment (INTERPANE) Coating suppliers (PRIREV); and End-users for air filtration and face masks (GVF); chemicals producers & industrial textiles (SIOEN). -Open Science and Data Management strategies in NanoBloc are targeting a widespread use of NanoBloc results by the scientific and industrial community to ensure a wider impact of the project outcomes. Public reusable and interoperable data regarding durability of the functionality of the coatings under different conditions are a key requirement in order to develop new Quantitative structure–activity relationships (QSAR) models[42] enabling Quality-by-Design fast developments in the future.
Provide business opportunities especially for SMEs	**- 3 technology and manufacturing SMEs are actively present in NanoBloc®**: GVF, PRIREV and IRIS. - Results obtained by NanoBloc® will be intensely disseminated **to reach other SMEs** interested in the application of the NanoBloc® coatings in their products. Pathways to SMEs will be established via outreach to materials and deep tech regional, national and European-level Associations and Clusters, such as the EuraMaterials cluster, European Advanced Textile Materials' Cluster, Catalan Advanced Textiles Cluster, among others.
Sustainable synthesis of nanocoatings (including bio-based materials) especially with effectiveness against a range of pathogens	- Technologies for the synthesis of nanocoatings/ nanocomposites in NanoBloc® such as PVD, is considered **one of the most environmentally friendly coating technologies**, due to the lack of waste products, to the "dry" process characteristics (no liquid involved) and to the low consumption of starting materials (coatings are few nanometers thick). -Wet chemistry: we will develop and/or scale up new synthesis (including bio-based materials) with specific focus on process sustainability (solvent-free processes).
Research areas should address new antiviral and antibacterial nanocoatings for a range of applications on both surfaces of high-traffic objects (door & window handles in public places, public transport, hospitals, public buildings, elderly homes etc) & coatings for textiles (protective clothing […], laboratory coats, face masks, etc.)	- New antiviral & antibacterial nanocoatings for a range of applications addressing use on both surfaces of so called high-traffic objects and coatings for textiles based on POLITO Ag/silica co-sputtered coating (published results), PRIREV & TEKNIKER Cu coatings are **inorganic nanomaterial based coatings with demonstrated and enhanced antimicrobial and antiviral activity.** **- NanoBloc® will push toward the highest TRL the co-sputtered Ag/silica & Cu based coatings and biobased (chitosan) antiviral and antibacterial nanocoatings** - Prototypes will be provided, tested & demonstrated to TRL6 **in working environments** for **high-traffic objects: door handles**, sanitaryware-taps, etc. (PRIREV) and coatings **for textiles** such as mattresses, **protective clothing**, coatings for wallpaper & furniture (SIOEN) **and air filtration and face masks** (GVF). Validation in real but controlled environments in the project (Hospital & shared facilities in offices) will open the path to future applications **in public places**, (public transport & buildings, hospitals, schools, elderly homes) and other applications as food processing plants and others.

2. Wider impacts, in the longer term, specified in the respective destinations in the work programme

[42] Ambure et al. QSAR-Co: An open source software for developing robust multitasking or multitarget classification-based QSAR models. *J Chem Inf Mod* 59.6 (2019):2538-2544.

NanoBloc® will make a clear contribution to the destination described in the topic: **'Increased autonomy in key strategic value chains for resilient industry'**- NanoBloc will contribute to **advanced materials value chains** that are high performance, safe, and sustainable 'by design', contributing both health and wellbeing benefits for society (reduced spread of bacterial/viral/fungal infections), as well as contributing to meet the challenges of climate neutrality, transition to a circular economy and a zero-pollution Europe (as described in Table 9 in Section 1.2). In particular, NanoBloc® contributes with a credible pathway to expected impact of Cluster 4: *'Global leadership in clean and climate-neutral industrial value chains, circular economy and climate-neutral digital systems and infrastructures* (networks, data centres), through **innovative production and manufacturing processes and their digitisation**, new business models, **sustainable-by-design advanced materials and technologies** enabling the switch to decarbonisation in all major emitting industrial sectors, including green digital technologies.'

In particular, and as outlined in the Destination theme in the work programme topic, **the COVID-19 crisis has shown that global competitiveness and resilience are two sides of the same coin**. Considering viral infections alone, the world has encountered a number of viral pandemics in recent history (viral infections such as SARS, MERS, and the ongoing COVID-19) that have caused tremendous morbidity and fatality, and morbidity among people and communities. NanoBloc® will help Europe to be better equipped to **withstand and cope with shocks**, thanks to scientifically validated antiviral, antibacterial and antifungal nanocoatings that can prevent the spread of infections, using technology that is green, durable and versatile to ensure wide uptake. Moreover, by being able to extend the durable life of protective clothing (such as masks and gowns) we will contribute environmental, cost and supply security benefits. Common surfaces are used regularly by the public and therefore can be repeatedly contaminated following each use. **Disinfection of contaminated surfaces using bleaches or similar chemicals** requires frequent repetition, making it an expensive process that also releases irritant gases (certain disinfectants can also trigger asthma and can be linked to other chronic respiratory conditions[43]), and generates potentially unsafe disposal of contaminated plastics and containers after use. It is imperative on the longer term to be able **shift to the large scale use of self-cleaning surfaces in our everyday life**. As most of the available antiviral surfaces only work temporarily, our focus is the development of surfaces with **long-term stability and durability**.

Thanks to a relevant research-industry cocreation, our NanoBloc® project includes the scientific, technological and commercial actors needed to codesign and provide the key technology and market pathways to '**New sustainable-by-design** *[antiviral, antibacterial and antifungal nanocoating]* **materials with enhanced functionalities** *[reliable, robust, durability]* **and applications in a wide range of industrial processes** *[PVD, EPD, sol-gel, UV-curable coatings]* **and consumer products** *[face masks, air filtration systems, clothing, matresses, wallpaper, furniture, door knobs, handrails, sanitaryware]*.**'**

Likewise, we provide a pathway to European **'Leadership** *[based on new scientific validation standards for assessing performance of biocide nanocoatings for guiding consumers, manufacturers, researchers and policy makers]* **in producing materials that provide clean, toxic/pollutant free environment solutions***[self-cleaning surfaces based on a green technology that uses inert gasses, metal and ceramic bulks, nor any chemical precursor or solvents or waste critical solutions]***, decarbonising industry, and safeguarding civil infrastructures.**

We will also contribute to **'Leadership in circular economy** *[use of bio-based materials, e.g. chitosan]* **that strengthens cross-sectorial cooperation along the value chain and enable SMEs to transform their activities and business models.',** as well as to **'Increased adoption of key digital and enabling technologies** *[Photonics and Artificial Intelligence based Digital Twin and real-time process monitoring technology]* **in industrial value chains and strategic sectors, paying particular attention to SMEs and start-ups** *[3 SMEs in our consortium and we will create new commercial opportunities for a wider SME-base including startups]***.'**

Target groups that will benefit

The target groups that will benefit from the outputs/results of our project are summarized in Table 14:

Table 14- Target groups of the NanoBloc® project

Target Group	Benefit
Coating manufacturers & surface treatment companies	Scaled up and validated technologies and processes
Medical and healthcare systems	New products that are coated with tested and validated antibacterial, antimicrobial and antifungal properties with outstanding performance to protect patients and healthcare workers from infection & increase resilience of our healthcare systems.

[43] Dumas et al..Association of occupational exposure to disinfectants with incidence of chronic obstructive pulmonary disease among US female nurses," JAMA Network Open 2(10), (2019)

End-users (incl. SMEs): e.g. indoor hygiene (retail, hotels, workplaces, homes…), sanitary, textiles, etc.	New products that are coated with tested and validated antibacterial, antimicrobial and antifungal properties with outstanding performance. Greater resilience, less impact from sick leave impacts to their workforce.
Consumers	Science-based claims about the virucidal and biocidal effectiveness, duration, safe and environmentally friendly properties of the new protective products
Standardisation bodies	Scientific evidence to inform new standards relevant for virucidal testing.
Scientific/Research community (field of nanocoatings)	New knowledge in relation to improved testing methods for validating R&D in virucidal effectiveness
Society, Healthcare workers, Patients, Citizens & governmental stakeholders	According to a study published in 'Science, Ecology and Economics for Pandemic Prevention' some economists predict that the economic & mortality costs of responding to the COVID-19 pandemic could reach $10-20 trillion. The global adoption of the NanoBloc® coatings for air filtration & microbe /virus free surfaces will help reduce risk of infection and reduce such devastating impacts in the future.

b. Requirements & barriers that may determine whether the desired outcomes and impacts are achieved

Table 15 summarises the requirements and potential barriers that could determine the achievement of the desired outcomes and impacts of the project and provides mitigating measures in the case of each.

Table 15- Requirements/Barriers and Mitigating measures

Requirements/ Barriers	**Mitigating measures**
Other R&I work in & beyond HE	Continuous literature/patent update, technology watch, participation to conferences and fairs, in order to implement mitigation strategies to avoid duplication and to foster collaboration.
Regulatory & standards requirements	Regulatory issues related to validation of the proposed nanocoatings will be duly assessed during via the cocreation work that will be performed with industry in Task 4.1 Cocreation for an application-oriented development, as well as in WP5- Validation and Impact. We have a regulatory partner (UNE) in our consortium: their role is to address regulatory issues related to validation of the proposed nanocoatings. In the event of regulation prohibiting e.g. metal nanoclusters, NanoBloc® proposes a wide range of coatings and we have a range of alternatives available. On the enabling side however, a tightening of standards in relation to e.g. antiviral validation (which could be promoted by NanoBloc scientific evidence and protocols would give us a competitive advantage.
User behaviour & social norms towards preventing infections	Given the current Covid-19 situation, we have seen a marked shift in user behavior towards new social norms (e.g. majority of the population protecting themselves and others against the virus). With further pandemics and health risks foreseen by WHO, it seems likely that face masks and in general protections against bacteria, fungi and viruses are here to stay and will form part of the New Normal. Indeed, from the start of this pandemic, there was a dramatic drop in flu & other seasonal diseases[44,45].
Existing commercial products	There is a vast number of available commercial products claiming their effectiveness against viruses & bacteria, which rapidly came on the market to service urgent demand due the pandemic. As shown in section 1.2, the claims made by these products have not been sufficiently scientifically validated. NanoBloc® aims to develop a scientifically founded and versatile platform of antimicrobial/virucidal/antifungal nanocoatings, with scientifically validated and demonstrated effectiveness, stability and environmental friendliness, for a wide range of innovative applications by a world-class European consortium.

c. Scale and significance of the project's contribution to the expected outcomes and impacts

Key assumptions based on current knowledge and baseline for NanoBloc expected outcomes

The European Centre for Disease Prevention and Control (ECDPC) estimates that HAI (health care-associated infection) burden in Europe has significant health & economic impacts, affecting >4.5 million people each year, leading to meaning 16 million extra days of hospital stay at a direct cost of €7 billion/annum and €30 billion associated indirect costs[46]. Moreover, 147000 deaths/annum in EU are attributable to HAIs (directly or contributing). One of the main concerns is Antimicrobial Resistance (AMR) related to antibiotic resistance developed by pathogens that makes it necessary to apply different strategies to reduce HAIs[47,48].

[44] https://www.hsph.harvard.edu/news/hsph-in-the-news/a-sharp-drop-in-flu-cases-during-covid-19-pandemic/
[45] https://www.the-scientist.com/news-opinion/the-pandemic-crushed-the-flu-what-happens-when-it-returns-68951
[46] Murphy et all Reduction of HAIs with antimicrobial inorganic nanoparticles incorporated in medical textiles: an economic assessment." *Nanomaterials* 10.5 (2020): 999.
[47] https://www.ecdc.europa.eu/en/news-events/ecdc-calls-continued-action-address-antimicrobial-resistance-healthcare-settings
[48] Hall, Thomas J., et al. "A call for action to the biomaterial community to tackle antimicrobial resistance." Biomaterials Science 8.18 (2020): 4951-4974.

Although there is laboratory evidence of effectivity of biocidal coatings against a range of pathogens, there are little data available about effectiveness of antibacterial coatings in healthcare environments and potential reduction of HAIs[49]. Studies on copper surfaces[50] on real environments account for up to 80% of reduction of bacteria compared to control surfaces in real environments and leading a 50% reduction in the risk of acquiring HAIs[51], by placing antibacterial copper surfaces in high-traffic surfaces on specific areas of Intensive Care Units. Nevertheless, other studies point that only 30% transmission events are environmental, including air[52]. Specific studies for antibacterial textiles in healthcare environment provide values between 20-40% reduction of HAIs. On the basis of these data, and taking into account that NanoBloc targets all high traffic objects, textiles and air filtration, we can assume 3 scenarios by using biocidal surfaces on specific sites in healthcare environments: Conservative (20% reduction of HAIs), (Medium: 35% HAIs reduction) and Optimistic (50% HAIs reduction). Table 16 shows how widespread adoption of biocidal surfaces **for bacteria alone** could contribute to a reduction of HAIs, deaths and costs.

Table 16- Reduction of HAIs, associated deaths and costs due to adoption and use of biocidal surfaces

Scenario	Affected people millions /per year	Deaths	Cost (€ billion)
Current	4.5	147000 (direct + indirect)	37 (direct + indirect)
conservative reduction of HAIs	0.9 reduction	29400 reduction	7.4 savings
medium reduction of HAIs	1.57 reduction	51450 reduction	12.95 savings
optimistic reduction of HAIs	2.25 reduction	78500 reduction	18.5 savings

The ECDPC has so far reported **36,307,572 COVID19 cases and 750,921 deaths in the EU**[53]**.** The number hospitalizations needs to be estimated as reporting differs in different countries: taking as a baseline the 2nd and 3rd waves peaks in Spain[54], the cases/hospitalizations rate was around 7.5% and hospitalization/deaths about 20%. Studies on nosocomial COVID19 infections are scarce and differ[55] but ranges between 44% and 0.1%[56] (1% baseline assumed for calculation) with 30% of nosocomial COVID19 infections ending in death. Studies also differ on COVID19 hospitalized people acquires a bacterial/fungi HAI during their stay in the hospital, but data ranges from 50%[57] to 8%[58] with 5 times higher mortality in HAI affected patients. **That would mean around 65,000 deaths occurring in Europe in healthcare environments due to nosocomial COVID19 or combined with other HAIs.** There are no data about antiviral surfaces efficacy on healthcare environment but assuming similar scenarios as in bacteria, that would mean between 13,000 and 32,500 lives of COVID19 European patients saved if we had biocidal surfaces to prevent nosocomial infections against virus and bacteria in healthcare environments alone.

While it is impossible to estimate the overall impact of such biocidal surfaces being also available in nursing homes, public transport, schools and other public places, rough figures for healthcare environments alone give an initial quantitative idea on the potential impact of widespread adoption of effective biocidal surfaces in the healthcare system. With the above mentioned **assumptions between 40,000 and 110,000 deaths could have been prevented in Europe since the beginning of the pandemic, while direct reduction of workforce loss preventing illness and hospitalizations attributed to HAIs could also be calculated.**

Scale of NanoBloc impact and outcomes

Table 17- Direct scale/capacity of the NanoBloc partners

Application	Partner	Scale brought by the NanoBloc partners
Textiles and air filters (R2R processing)	SIOEN	World market leader in coated textiles with 60+% EU market share on coated textiles for e.g. truck tarpaulins and a production capacity of 100+ million square meters/per year. Specifically on medical textiles (mattress covers & aprons) provide 5-8 mio m² to the market. This corresponds with ~30% of global market and ~50% of European market.
	GVF	Current capability of 200k filter elements intended for air treatment. Main markets are Europe, Middle East and Russia
	INTERPANE	INTERPANE is currently manufacturing equipment for large area coating and for glass, are coating about 50 million m² per annum worldwide.
High Traffic objects (Batch	PRIREV	Currently has 9 PVD chambers and are providing coatings for aesthetics purposes on parts such as sanitary taps and door handles to different countries in the European market with capability to coat 3000 sanitary water taps or door handles per day while

[49] https://www.hygiene-in-practice.com/antimicrobial-coatings-a-new-opportunity-for-reducing-nosocomial-infections/
[50] CHYDERIOTIS S et al. New insights on antimicrobial efficacy of copper surfaces in healthcare environment:a systematic review. Clin Microbiol Inf 2018,24(11) 1130-1138
[51] Salgado et al. "Copper surfaces reduce the rate of healthcare-acquired infections in the intensive care unit." Infection Control & Hospital Epidemiology 34.5 (2013): 479-486.
[52] Danceret al. Tracking Staphylococcus aureus in the intensive care unit using whole-genome sequencing. J Hosp Infect 2019; 103(1):13-20
[53] https://www.ecdc.europa.eu/en/cases-2019-ncov-eueea. Data at 31/08/2021.
[54] https://www.mscbs.gob.es/profesionales/saludPublica/ccayes/alertasActual/nCov/situacionActual.htm
[55] Baccolini et al. The impact of the COVID-19 pandemic on HAIs in intensive care unit patients: a retrospective cohort study." Antimicrob Res & Inf Con 10.1 (2021): 1-9.
[56] Rhee, Chanu, et al. "Incidence of nosocomial COVID-19 in patients hospitalized at a large US academic medical center." *JAMA network open* 3.9 (2020): e2020498-e2020498.
[57] Cox, Michael J., et al. "Co-infections: potentially lethal and unexplored in COVID-19." The Lancet Microbe 1.1 (2020): e11.
[58] Moreno-Torres, Víctor, et al. "Bacterial infections in patients hospitalized with COVID-19." Internal and Emergency Medicine (2021): 1-8.

processing)	TEK-NIKER	Has industrial chamber to add 2500 door handles/day. Previous experience transferring turnkey solutions for decorative coatings on door handles delivered capacity to coat around 5000 door handles/day < 1 € additional cost for the coating.
	SIOEN	Applications in the sanitary flooring and furniture markets are estimated to 250 T coating per year (at approx.10 €/kg)

In 2018 there were 2.4 million hospital beds[59] in EU distributed across some 12500 hospitals[60]. Although no data is available on square meters consumption of hospital textiles, reference data on daily dirty laundry weight/bed[61] can provide an indication of the amount of non-disposable textiles to be 18 m^2/bed/day with would translate to 86 million m^2 as the minimum textile surface required to cover all needs in Europe (assuming a minimum of one set of textiles in use while another one is in the laundry). If we take into account current capabilities on R2R processing of SIOEN and INTERPANE shown in table 17 could potentially cover a significant part of the overall demand in Europe.

The same is applicable for batch processing and high traffic objects. the supply required to provide each hospital in EU with door handles and sanitary taps tabs (baseline estimation: 2 beds/room; 2 doors/room, 2 handles/door, 2 sanitary taps/room) would be 4.8 million door handles and 2.4 million sanitary taps. Current PRIREV capability of supply is > 1 million door handles or tabs per year (to which TEKNIKER capabilities or licenses to third companies could be added), so taking into account that these solutions would be long-lasting the demand for all patient rooms in Europe could be covered by the NanoBloc consortium in <5 years.

Therefore, even with rough numbers, the consortium capabilities are enough to provide a wide significant impact. Even if it is not realistic thinking that all the market share will be dominated by NanoBloc partners or would dedicate all their capabilities to this single market, **it is clear that covering the potential demand in Europe is realistic, as it also assumes the growth and expansion of the NanoBloc companies to meet demand (generating increased turnover and jobs).** As previously mentioned, further supply outside the consortium could be achieved by licensing (and in particular creating opportunities for SMEs) or by taking advantage of new protocols and data from NanoBloc, but also by other technologies that will be up-scaled in NanoBloc, such EDP (no current production capabilities in consortium, but **this will provide new opportunities for European industry, with a particular focus on SMEs, as we pave out pathways to exploitation through** Task 6.4**)**

Significance of NanoBloc impact and outcomes

Key needs to adopt antimicrobial surfaces & technology adoption cycle: Penetration in the market is a key aspect for reaching the expected impact. Market expectations are good, as pandemics are forecast to drive demand for antimicrobial coatings in the coming years[62]: the global market is expected to duplicate from 2019 (>3.2 billion $) to 2026 (6.3 billion $). It is noteworthy that big shares of this market correspond to silver antimicrobial coatings, healthcare applications and indoor air quality markets; all are addressed by NanoBloc. Importantly, additional costs due to the coatings are lower than additional cleaning operations to guarantee disinfection as calculated by Schmidt et al.[63] for copper surfaces and even for biocide textiles when compared to expenditure on antiseptics[64].

In order to realize this potential and growth in the coming years and thereby yield real impact on health, it is vital that new products that are brought to market are solidly built in terms of performance. Although the potential of biocidal surfaces has been proven at laboratory scale through standard tests, professionals and researchers of healthcare systems clearly point to the need of having **more data on their real impact on relevant environments and their durability and resistance** specially to cleaning chemicals and to **avoid the need of reapplication** [65], as well as **risk assessment of potential toxicity and ecotoxicity**[66]. **All these needs are addressed in our project, so it is reasonable to foresee wide acceptance of the NanoBloc solutions.** Although, NanoBloc targets TRL6, the advantages of the cocreation approach, and up-scaling and demonstration activities should accelerate further steps upwards in the TRL scale after the termination of the project, towards product commercialization (which will be formalised in professional business plans that will be prepared in Task 6.4). The Nanobloc partners are committed to a competitive Time to Market (TTM)- 3 to 5 years, depending on the technology and product. Assuming typical normal distribution on innovation adoption[67], considering that Nanobloc partners are ready to cover a big share of demand in Europe in terms of production required, we can consider that **Innovators** (2.5%) could adopt NanoBloc solutions during the first 5 years after commercialization (expected year 8-10 after project termination), those experiences would help to build key dynamics on diffusion of innovation[68] such us demonstrate *Relative advantage*

[59] https://ec.europa.eu/eurostat/statistics-explained/index.php?title=Healthcare_resource_statistics_-_beds#Hospital_beds
[60] https://hospitalhealthcare.com/latest-issue-2020/hope/hospitals-in-europe-healthcare-data/
[61] Lopes, Catia Milena, et al. "Analysis of sustainability in hospital laundry: The social, environmental, and economic (cost) risks." *Resources* 8.1 (2019): 37.
[62] https://www.gminsights.com/industry-analysis/antimicrobial-coatings-market-report
[63] Schmidt et al. November 2019 Appl. Environ. Microbiol. doi:10.1128/AEM.01886-19 Copyright © 2019 American Society for Microbiology
[64] Murphy et al. Reduction of HAIs with antimicrobial inorganic nanoparticles incorporated in medical textiles: an economic assessment." *Nanomaterials* 10.5 (2020): 999.
[65] Ellingson et al. Impact of a Novel Antimicrobial Surface Coating on HCAI and Environmental Bioburden at 2 Urban Hospitals. *Clin Inf Dis* 71.8 (2020): 1807-1813.
[66] Ahonen, et al. "Proactive approach for safe use of antimicrobial coatings in healthcare settings: opinion of the COST action network AMiCI." *Int J Env Res Pub He* 14.4 (2017): 366.
[67] Balas, E. Andrew, and Wendy W. Chapman. "Road map for diffusion of innovation in health care." *Health Affairs* 37.2 (2018): 198-204.
[68] Cain, Mary, and Robert Mittman. "Diffusion of innovation in health care." (2002).

of the innovation, *Observability* of the solution permeating through *Homophilous groups* (i.e. those with similar characteristics), *Compatibility* to coexist with current systems and *Infrastructure* availability to implement the solutions. Thus, we can consider that **Early adopters** (a further 13.5%) could be covered after another 5 years, and **Early Majority** (additional 34%) by year 18-20 after finishing the project.

Under that assumptions and scale defined previously for EU, Table 18 shows estimations on NanoBloc potential impact, assuming 2030 as **TTM deadline**, from Conservative to Optimistic scenarios in terms of performance of the solutions preventing HAIs. Calculations were done only for non-pandemic scenario (typical bacteria and fungi HAIs): virus were not considered for the calculations as each potential pandemic could have their own characteristics and there is no way to preclude consistent data on real impact.

As many assumptions have been made, and due to uncertainties on some basic data, the figures can be considered as an estimation of the potential significance of the NanoBloc results and how this potential can be achievable on the basis of current production capabilities in the consortium and the scarce current knowledge of potential impact of biocide coatings in reducing HAIs in real environments. Moreover, these significance data calculations have been made considering only Healthcare systems (hospitals) without pandemic virus scenario or other targeted applications such as nursery homes, public transport or other public places that were not considered due to lack of consistent initial data for assumptions. As such, even if some of the mentioned figures could be overestimated, the overall potential impact will be significantly higher considering other applications.

Table 18- Estimated impact in EU of Nanobloc Outcomes

Scenario	Impact metrics	2035 Innovators 2.5%	2040 Early adopters 16%	2045 Early Majority 50%
Conservative (20% HAI reduction)	Patients HAIs affected avoided/ year	22,500	144,000	450,000
	Lives saved/ year	735	4704	14700
	€ billion saved on healthcare / year	0.185	1.184	3.7
Medium (35% HAI reduction)	Patients HAIs affected avoided/ year	39,250	251200	785,000
	Lives saved/ year	1286	8232	25,725
	€ billion saved on healthcare / year	0.323	2.072	6.475
Optimistic (50% HAI reduction)	Patients HAIs affected avoided/ year	56,250	360000	1,1125,000
	Lives saved/ year	1962	12560	39250
	€ billion saved on healthcare / year	0.465	2.96	9.25

2.2 Measures to maximise impact - Dissemination, exploitation and communication

First version of the 'Plan for the dissemination and exploitation including communication activities

We outline hereunder our planned measures to maximise the impact of our project through our initial '**Plan for the dissemination and exploitation, including communication activities'** (hereafter DE&C Plan). Due to the dynamic nature of EU collaborative projects (and also considering the time lapse between proposal preparation and implementation), **this first version of the Plan will be further detailed in Deliverable 6.2 (at M6)**. As a central element of our project methodology is the cocreation between the research and the industry partners to ensure that science meets industry requirements, our **DE&C Plan will be periodically updated according to the progress and emerging results of the project.**

Continuous results assessment and value proposition design: The results of our project will be considered as **any tangible or intangible output of the action, such as data, knowledge and information- whatever their form or nature, whether or not they can be protected- and which can create impact during and/or after the EU funding procedure**. An ongoing activity that will accompany the R&D work will be to continually assess and catalogue the project outputs as they emerge, and to consider the value and **potential unique selling points (USPs)** of the key results towards our different target groups.

Communication Plan and Toolbox: While the **specific details of the communication activities to be implemented will be appropriate with the nature of each of the final outputs of our project** and in line with the progress of the development of the action, we have defined a strategically planned, holistic approach that will govern the different ad-hoc efforts. First of all, we have defined **clear communication goals and objectives**:
To **reach out to society** as a whole- there is an important opportunity to inform the public of the research-industry effort being leveraged towards delivering innovative products to the market that will play a high value role in protecting the public against the spread of infectious diseases and viruses thanks to high performance and scientifically validated new antibacterial, antiviral and antifungal coatings. We will organize a series of

webinars/focus groups seminars targeted at Civil Society Organisations (consumer-oriented) and to a wider audience with open sessions for addressing questions and concerns of the citizens that can be taken into account in the cocreation circle. Well known initiatives such as, European Researcher nights, Pint of science festivals and/or Local TEDex talks will be contacted to promote talks on NanoBloc results on their events to reach a wider audience.

- To **establish synergies** with external bodies such as standardization bodies (& contribute to improved science-based standards), & national/international industry associations (reach adopters of NanoBloc® innovations)
- To achieve **high visibility through the scientific community**
- To show the relevance (in our current Covid-19 era) of our project with the pandemic and its value of contributing **to mitigate future risks and to demonstrate how EU funding contributes to tackling the societal challenge** in relation to our vulnerability in general to bacteria, mould and viruses, and how we can protect citizens from such risks, also highlighting socio-economic impacts, and environmental contribution.

We will design and build a **Communication Toolbox** for achieving our communication goals and objectives:

- **Visual identity and brand** (e.g. we have already worked on a project logo during the bid preparation phase, and during project implementation we will invest effort in building brand identity and particularly in relation to the association of **NanoBloc®** with trusted science-validated claims; we will work on a brand 'claim' or tag line that effectively communicates a trustworthy brand value)
- **Project website** as an important vehicle for the project Communication (and dissemination) effort, hosting project public deliverables and also hosting blog content (e.g. Thought Leadership articles and whitepaper (e.g. in relation to the importance of standards, the impacts of research-industry cocreation), at least 6 newsletters, etc., which will also be disseminated via our **social media channels** (LinkedIn and Twitter (whereby we will create campaigns for our different target audiences).
- Informative material in **printable** (banners, posters, flyers) **& multimedia** (videos, digital flyers) formats
- Engagement with industry- participation in at least 3 **tradeshows/exhibitions**, face2face meetings, mailings…
- Achieve at least 6 **commercial publications** in online/offline magazines and blogs.
- PR strategy- well timed and prepared **Press Releases to coincide with key development milestones and outputs** (issued through relevant press channels, including the European Commission News Section).

Dissemination of the Project Results: In line with our Open Science Strategy, we will ensure the early and **wide dissemination or public disclosure of all the non-confidential project results**, via appropriate means, including scientific publications. As such, our **Dissemination strategy will: Identify the results** to disseminate and **transfer the knowledge and results to the stakeholders** that can best make use of it; Identify the appropriate **dissemination channels; Maximize the impact** of research, whereby we enable the value of our results to be potentially wider than the original focus; and, **Monitor and evaluate** the effects of the dissemination activities**.**

The main **dissemination tools**, among others, that we will use are:

- **Scientific publications** (at least 15) in relevant journals- e.g. Journal of Coatings Technology and Research; Nano Convergence; Surface and Coatings Technology, Applied Surface Science
- Participate (presential & virtual attendance) in at least 10 **international conferences**- oral/poster/papers, e.g. World Congress on Materials Science & Engineering; 10th International Conference on Smart Materials & Nanotechnology; Annual ICACC -International Conference & Expo on Advanced Ceramics & Composites.
- **Share results on online media** (research data, software, reports)
- Organize at least 2 **training and workshop sessions**
- **Mid/Final project** event to demonstrate to potential users/customers.

Standardization activities: Standardization in R&I projects supports the translation of the R&I findings into marketable solutions. NANOBLOC will build upon existing standards during the development of the novel processes and validation protocols ensuring compatibility with market conditions and increasing transparency for prospective customers. NANOBLOC will be linked with ongoing standardization work like, e.g., in ISO/TC 229 'Nanotechnologies', ISO/TC 107 'Metallic and other inorganic coatings', ISO/TC 201 'Surface chemical analysis' and ISO/TC 194 'Biological and clinical evaluation of medical devices'. This will support the dissemination of the project and its results, getting feedback from the stakeholders organized in standardization and to contribute to the development of new standards by forwarding inputs from NANOBLOC developed protocols. UNE, representing CEN-CENELEC, will lead these activities.

Exploitation actions: The main goal of the exploitation plan is **to make use of the results for commercial/business, societal, scientific, contribution to standards & even political purposes** (e.g. value & impact of NanoBloc® for protecting against the future virus threats and building resilience). Envisaged exploitation actions include (and table 21 at the end of this section summarises the Exploitation intentions of each member of the consortium):

- **Developing, creating & marketing a product or process**, or creating and providing a service. The global COVID-19 crisis has greatly increased industry demand for antimicrobial & antiviral coatings, especially for high touch surfaces in healthcare centres and long-term care facilities, public transport, schools, hotels, offices, etc to reduce human exposure and mitigate the spread of infectious pathogens. A clear ambition of the project is to bring the NanoBloc® technologies to market.
- **Patent** publications (target for at least 3)
- Utilizing the project results in **further research activities** other than those covered by the concerned project (or use of the results for academic purposes (PhD, post-PhD...)
- Contribution to new standards- (as discussed above).

Business Model and Go-to-Market strategy: We have used the **Business Model Canvas (BMC)** approach to focus our draft plan of the dissemination and exploitation of the outputs of the NANOBLOC® project. This first version of the BMC (Table 19) is a common Canvas for the project, however in Task 6.4 **each of the industry partners (INTERPANE, PRIREV, IRIS, GVF, SIOEN) will develop their own BMC based on the results that they will be commercialising or adopting**. These BMCs will be delivered at Project Interim and in the latter phase of the project they will be further elaborated into Business Plans that will be delivered at the end of the project.

Table 19- Business Model Canvas

Key Partners	Key Activities	Value Proposition	Customer Relationships	Customer Segments
Scientific knowledge & innovation providers: • POLITO • UNITO • FAU • TEKNIKER Tech & engineering industry providers: • INTERPANE • PRIREV • IRIS End-user industries (early adopters): • GVF • SIOEN Standardization: UNE	• Coating synthesis & characterization • Process scale up and optimization • Validation • Communication & Dissemination • Business planning for route to market • Fundraising for TRL7-9 activities and market deployment • Contribution to new standards **Key Resources** • Patent • NanoBloc trademark (akin to quality assurance for the market) • Scientific validations (key marketing tool) • Process & laboratory equipment • Process knowledge encapsulated in recipes and Digital Twins	Versatile & industrially scalable antimicrobial, antiviral and antifungal Nanobloc© nanocoating technology with superior properties: • Efficacy • Durability • Eco-friendliness That have been scientifically validated (market differentiation): • Robust marketing claims • Consumer confidence	• Cocreation cycle • Technology partnerships • Robust science-based marketing claims **Channels** • Website • Outbound marketing (direct sales teams in the industry partners) • Inbound marketing (content marketing, SEO) • Exhibitions & Conferences • Distributors • Networks (EU partnerships)	• Coating manufacturers and surface treatment companies • End user industries: textiles (masks and protective clothing-PPE, aprons, work overcoats, mattresses, etc.), sanitary, interiors/high traffic surfaces (e.g., hospitals and healthcare facilities, workplace, business environments, universities, schools and public environments, transport, retail, hotels and catering, households), indoor hygiene, medical & healthcare, plastics packaging etc.

Cost Structure	Revenue Streams
• Staff + overhead • COGs • Equipment & plant • Sales and Market costs • Channel partnership costs (e.g., distributor %)	• Sale of Nanobloc coatings • Sale of Nanobloc coated products (air filters, masks, aprons, mattresses, wallpaper…) • Sale of process equipment • Business opportunities in Industry 4.0 implementation- digital twins & process monitoring equipment

Strategy for IP Management: Our strategy for the management of intellectual property (IP) to ensure **Freedom to Operate (FTO)** and to adequately protect our results as relevant to ensure the commercial exploitation of the project. We have conducted **a first patentability study** (using patent libraries- EPO and WIPO) to identify potential IPR issues with existing technologies and patents, and we have established our FTO and differentiation points. **We have a clear understanding and agreement among the partners in relation to Background IP- ownership of Background remains with the current owners and is not affected by the project** & **access rights to Background needed to implement the project is provided royalty free, and if needed for exploitation of the project results it will be either royalty free or under fair and reasonable conditions.** Each partner has declared their **Background IPR** (will be formalised in the Consortium Agreement) and necessary access rights to e.g. Polito

patent (PCT/IB2018/057639- Novel technique for the deposition of a nanostructured coating with virucidal, antimicrobial and antifungal properties via co-sputtering technique). No Background IPR is needed from any Third Parties external to the consortium. Our strategy for knowledge management and protection is **to file new patents that will come as a result of the project, with a clear ownership strategy- developers own their results, and joint development will result in jointly owned patents**. If one of the parties participating in the project wishes to proceed with the protection of the results through a patent or other forms of industrial property rights, the others, if not interested in co-ownership, will transfer their share of ownership of the patent under conditions to be defined on a case-by-case basis. We will also follow other protection paths as appropriate, such as trademarks, copyright, Knowhow, Trade Secrets, etc.Table 20 **summarises the key knowledge items that we expect will be generated during the project, the foreseen protection measures**, the owner(s) and access rights to this key knowledge (to be included in a consortium agreement that will be generated and signed by all partners if our project is selected for funding, and the final Results Ownership List will be updated in the Final Periodic Report).

Table 20- Summary of the IP strategy

Knowledge Item	Protection	Owner(s)	Access Rights
New methodologies & protocols for study of the biocide mechanism of action	Open Access	UNITO TEKNIKER	Open Access: Disseminated in peer-reviewed journals & open databases
New processes & coating compositions for maximizing adhesion, durability, biocidal effect, specific for application/product	New patent or Trade secret	POLITO TEKNIKER other partners	Preferential access to partner end users & licensing outside consortium
Process parameters of co-sputtering coatings, transferred to upscaling facilities	New patent or Trade secret	POLITO INTERPANE	End users
New compositions and process parameters of RF co-sputtered coatings	New patent(s)	POLITO	End users
Process parameters of Sputtering & Cathodic arc of Ag doped nitrides	New patent or Trade secret	PRIREV	Licensing to Third Parties
Process parameters for PVD Cu doped nitrides and oxides	Open access	TEKNIKER	Open Access: Disseminated in peer-reviewed journals & open databases
Process parameters of Nanoparticle doped UV lacquers	New patent or Trade secret	SIOEN POLITO	Own commercialisation and Licensing to Third Parties
Process parameters of electrophoretic deposition of biobased formulations	Open Access	FAU	Open Access: Disseminated in peer-reviewed journals & open databases
Process parameters of Sol-Gel hybrid coatings	Open Access	FAU TEKNIKER	Open Access: Disseminated in peer-reviewed journals & open databases
Nanoparticle doped UV-lacquer with self-disinfecting properties	EU patent	SIOEN	Under license
New knowledge on Toxicity and Ecotoxicity of biocide formulations	Open Access	UNITO TEKNIKER	Open Access: Disseminated in peer-reviewed journals & open databases
Real-time PAT monitoring solutions and Digital Twins of the processes	Knowhow in software code	IRIS	Preferential access to process owners

Exploitation intentions of the consortium: Table 21 summarises the core exploitation intentions of each partners.

Table 21- Exploitation Intentions of the partners

Partner	Exploitation Intentions
POLITO	New publications on reputable international scientific journals: New knowledge/know how about antiviral /antimicrobial/antifungal coatings aimed at: students for updating the lectures contents; researchers for new collaborations; industrial partners for new collaborations and funded projects
TEK-NIKER	New publications on international scientific journals. IP Protection (patenting coatings and/or processes and industrial secret). Technology transfer to industry: industrial projects for different applications and environments (ceramic tiles, automotive, implants, food processing plants). Licensing agreements & coating services.
UNITO	Dissemination of the results to the scientific community and to operators in the sector of antimicrobial materials through publications in peer-reviewed scientific journals and presentations at specialized congresses in this field. The knowledge generated with the project will be also employed for updating the lecture contents for students and for the search of new scientific collaborations and funds
FAU	Development of antiviral & antibacterial nanocoatings by EPD and sol-gel method and their characterization
GVF	Effective application of a virucidal, antimicrobial and antifungal coating on filtration systems hence improving products quality and reliability.
SIOEN	To market coated textiles (incl. protective clothing, mattresses, and mouth masks), radiation-curable lacquers and varnishes for high-traffic objects & surfaces.
PRIREV	Upscaling of antibacterial/antiviral PVD films for high-traffic objects. Commercialization of the coating service for high traffic objects producers.
IRIS	Real-time nano-characterization Process Analytical Technologies (PAT) inline device to be added to our portfolio of photonics analysers for commercialisation, in addition to new AI models that can be integrated into our Scadalytics platform giving us access to a new client base in nanocoating production and application. Post-project we will continue work towards system prototype demonstration in operational

	environment (TRL7), progressing to a complete and qualified system (TRL8) for which we can commence commercialisation.
INTERPANE	Post project scale-up, market deployment and commercialization of the industrial coating equipment for the final users of the developed coating.
UNE	Participation of UNE as standards body intends to increase the exploitation possibilities for the other partners by using the benefits of standardization. This will reinforce the relevance of the European standardization system and its recognition as an effective exploitation channel towards the European R&I community.

2.3 Summary

SPECIFIC NEEDS	EXPECTED RESULTS	D&E&C MEASURES
• Covid-pandemic has triggered demand for **new antiviral and antibacterial nanocoatings** that offer **virucidal effectivity**, **extended durability** during use, are **safe and environmentally friendly**, and are suited **for a range of applications** addressing use in air filtration systems, in porous textiles (e.g. protective clothing in food processing plants, laboratory coats, face masks, etc.), and on surfaces of high-traffic objects (e.g. door and window handles in public places, public transport, hospitals, public buildings, schools, elderly homes etc). • Current testing standards are not enough to guarantee durability or functionality on real application environments. This has led to numerous products reaching the market since the covid pandemic making claims that are not fully supported by science-based testing and validation methods.	• Reliable Antimicrobial, Antiviral, and Antifungal Nanocoatings that respond to market needs in terms of **efficiency, durability and sustainability/ecofriendly properties**, with claims that are validated and backed by solid science-based evidence on antimicrobial and virucidal mechanisms of nanocoatings, to build resilience against future global health crises. • **Scaled up processes of PVD, Wet Chemistry and Organic/Inorganic nanocomposite coatings** • **AI-augmented PAT process monitoring and control platform** for facilitating process understanding and quality control • **Demonstrator products:** NanoBloc® porous filter materials (air filtration units, face masks), textiles (protective clothing, mattress covers, aprons, wallpaper), and high-traffic solid surfaces (doorknobs, handles, handrails, sanitaryware-taps, etc.), **the virucidal effectivity and durability of which have been proven with new fit-for-purpose testing protocols**	**Dissemination towards the scientific community, industry and health care environments**: Scientific publications with laboratory results and demonstration trials on relevant environments. Dissemination of the protocols to relevant standardization work groups. Participating in conferences and fairs; Participation in EC project portfolios to disseminate the results as part of a group and maximise the visibility vis-à-vis companies. **Exploitation of the new product:** Patenting the antimicrobial antiviral new products: air filters (GVF) and, textiles (SIOEN) and new coatings equipment and technology (INTERPANE). Patenting new coatings for high traffic objects (PRIREV, TEKNIKER). Direct exploitation of coatings services (PRIREV). Direct commercialisation of Real-time Process Monitoring and Digital Twins to support commercial adoption of the NanoBloc® technologies (IRIS). **Communication towards citizens:** Specific communication publications for wider audience. Participation on events such as Researchers Nights, Pint of science, and TEDex. Specific events for general public to show how the outcomes of the action are relevant to our everyday lives.
TARGET GROUP	**OUTCOMES**	**IMPACTS**
Coating manufacturers and surface treatment companies (access to innovative coating technology that is founded on a deep scientific base) **End users industries** for NanoBloc® antimicrobial, antiviral, and antifungal nanocoatings, which include: textiles (masks and protective clothing-PPE, aprons, work overcoats, mattresses, etc.), sanitary, interiors/high traffic surfaces (e.g. household, retails, hotels, workplace, business environments), indoor hygiene, medical & healthcare, plastics packaging etc. **Research/Scientific community** (field of nanocoatings) **Standards bodies** (new scientific-evidence to inform improved standards) **Healthcare workers, Patients, Citizens** (less risk of catching infections) **Business community and workplaces** (greater resilience, less impact from sick leave impacts to their workforce)	**Scaled up processes that can be applied in industry**: • **Batch processing:** 3D applications: PVD sputtering/cathodic arc/HIPIMS (PRIREV-TEKNIKER); Sol-Gel (TEKNIKER); EDP (FAU-TEKNIKER) • **Roll-to roll processing:** 2D applications (PVD Sputtering (INTERPANE); Sol-Gel (TEKNIKER); Nanocomposite coatings & lacquers (SIOEN) Business plans for the development of at least 9 new 'NanoBloc® protected' product lines- air filtration systems (GVF), face masks, mattresses, wallpaper, aprons (SIOEN), coating of doorknobs, taps and sanitaryware, handrails (adoption by PRIREV's client-base). New knowledge on virucidal and biocidal mechanisms for informing new and improved standards (engagement with standardization bodies will be commenced) High use of the new scientific knowledge (measured with the relative rate of citation index of project publications)	**Scientific:** New knowledge on virucidal and biocidal mechanisms **Economic:** Increased resilience to the spread of infections leading to less sick leave, reduced healthcare costs, rescued confinements, all of which are devastating to the economy. **Technological:** more effective, versatile and reliable antimicrobial/virucidal coatings respect vs. what is on the market; high temperature resistant antimicrobial/virucidal coatings for thermal re-generation of filters and other coated surfaces **Societal:** protection of citizen health and wellbeing against the spread of harmful viruses, bacteria and fungal infections (reduced incidences of aerosol and fomite infections). Increased consumer confidence in product claims that are backed by robust science.

3. Quality and efficiency of the implementation

3.1 Work plan and resources

Brief presentation of the overall structure of the work plan

Our workplan has been designed to coherently enable us over the course of 3 years to **develop, test, optimize, scale-up and validate (from TLR3 to TRL6), innovative, high performance new antiviral, antibacterial and antifungal coatings**. These **industrially scalable, versatile and green technologies are suitable for application on a variety of different substrates**: porous filter materials (air filtration units, face masks), textiles (protective clothing, mattress covers, aprons, wallpaper), and high-traffic solid surfaces (doorknobs, handles, handrails, sanitaryware-taps, etc.). We will also **plan and prepare for the post-project impact pathways**.

The workplan is divided into **7 inter-related work packages**, each with its own tasks delivering outputs (deliverables) and culminating in a series of milestones that will establish the project critical path and will help facilitate project monitoring and management. In Fig. 9 an overview of the development loops in the 3 years of the technical activities (or WPs) is shown.

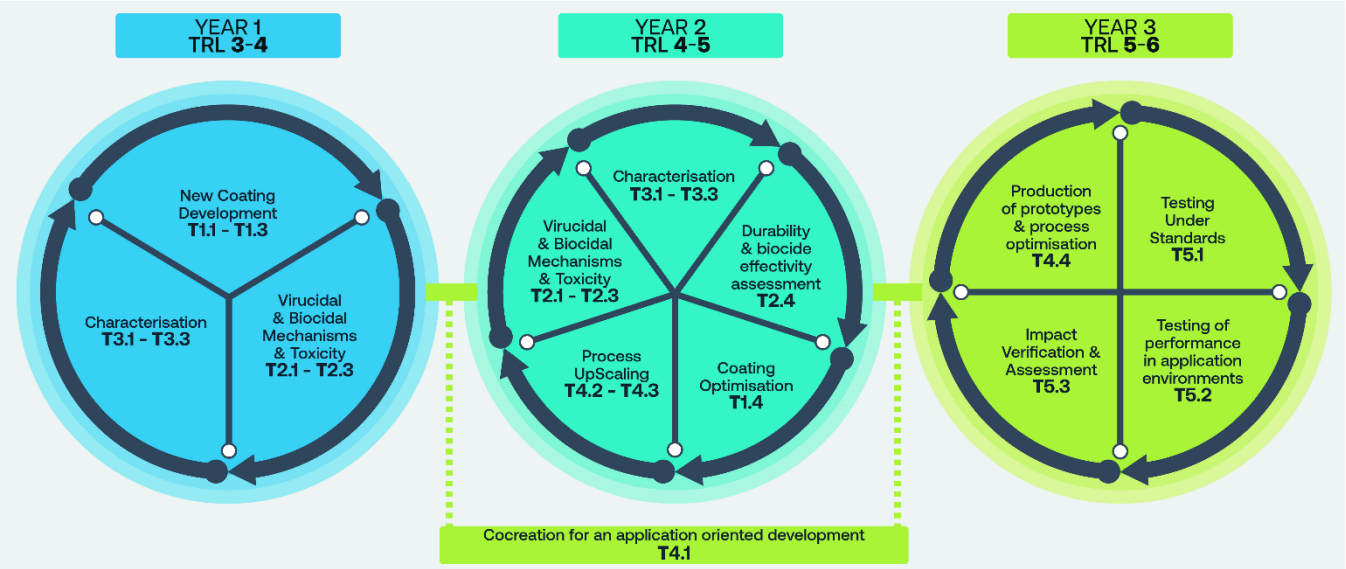


Figure 9- Development loops to advance the TRLs

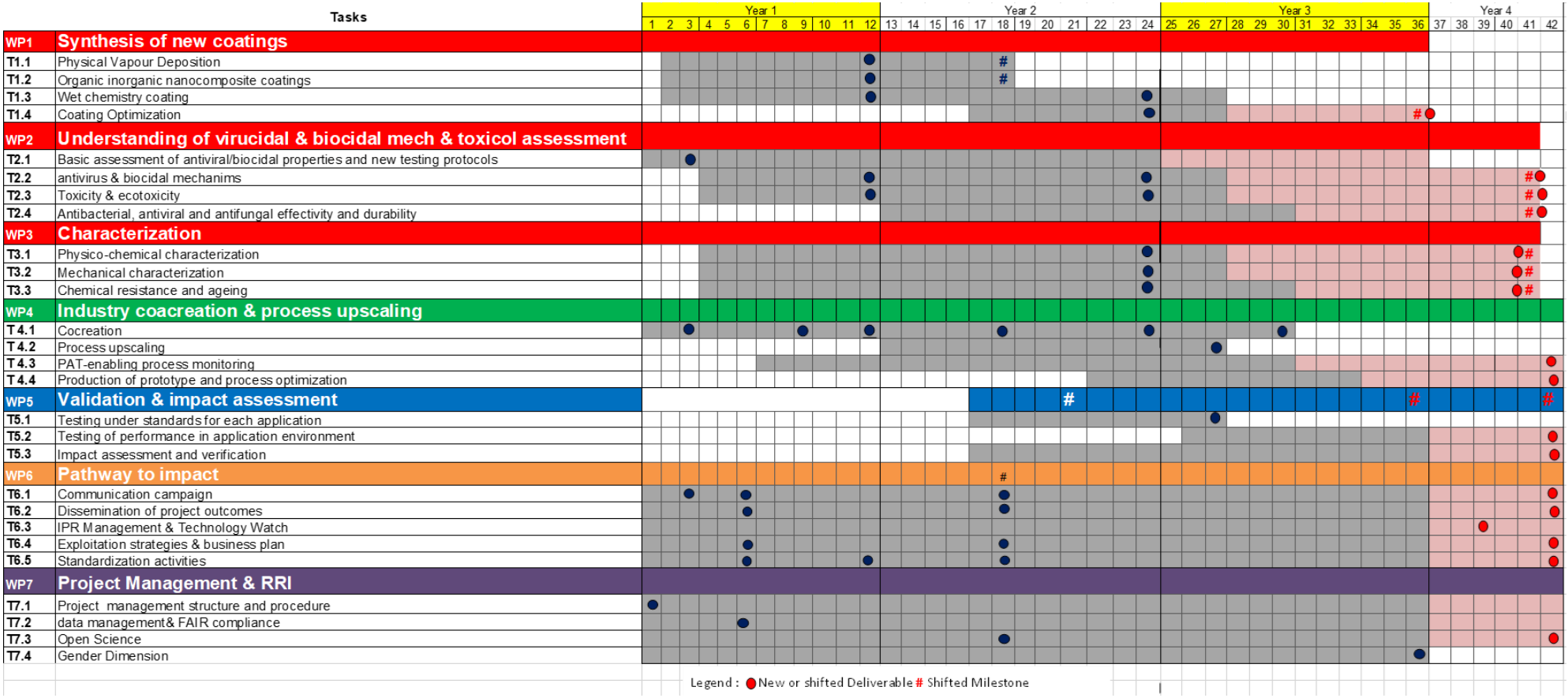
In year 1 of the project, in ***WP1 (Synthesis of new coatings)*** and through Tasks 1.1- 1.3, we will develop coatings by a) RF co-sputtering and other Physical Vapour Deposition (PVD) such as arc evaporation and combination of them b) Doping UV cured lacquers and varnishes with nanoparticles, and c) through other promising coating technologies, such as sol-gel, Electrophoretic Deposition (EPD) on substrates selected by end users and the development process will be codesigned and guided in close conjunction with the industry upscalers and end-users from Task 4.1 *(Cocreation for an application-oriented development)*. In order to verify the behaviour of the coatings, through ***WP2 (Understanding of virucidal & biocidal mechanisms and toxicology assessment)*** we will perform assessment the antiviral & biocidal properties, toxicity & ecotoxicity of the newly developed coatings, as well as characterize their physico-chemical, mechanical, chemical resistance and ageing performance through ***WP3 (Characterisation)*** in a complete development loop.

In year 2, once we have the first loops of development with WP2 and WP3, we will start with T4.2 up-scaling in ***WP4 (Industry Cocreation and Process Upscaling)***. We are looping here T1.4 (coating optimization), T2.4 (we will also be able at this point to combine durability and biocide effectivity assessment), T4.2 (process upscaling), as well as standards testing (T5.1), to fine tune the coatings formulations & parameters taking into account the results from durability+effect (T2.4) and results from up-scaling in which some problems could arise that may require optimization, and also with data from the machine learning activities from T4.3 (PAT enabled process monitoring and digital twinning).

In year 3, we will continue with the new optimization in task 1.4 and on process upscaling in T4.2, as well as testing under standards for each application, **testing of performance in application environments**, as well as impact assessment & verification ***(WP5-Validation)***.

We have depicted these development loops in the schematic above (Figure 9). Cross-cutting the workplan, we will perform comprehensive activities to deliver a ***Pathway to Impact (WP6)*** for the project outputs via a communication and dissemination of the project outputs, IP management, as well as business plan and defining a post-project plan for further scale-up towards market entry of the innovations. Finally, we will ensure a well-run project through ***WP7- Project Management & RRI***, which encompasses, in addition to project monitoring, tracking and risk management, data management, management of Open Science, as well as Inclusiveness and Gender Dimension.

Timing of the different work packages and their components (Gantt chart or similar)



Graphical presentation of the components showing how they interrelate (Pert chart or similar)

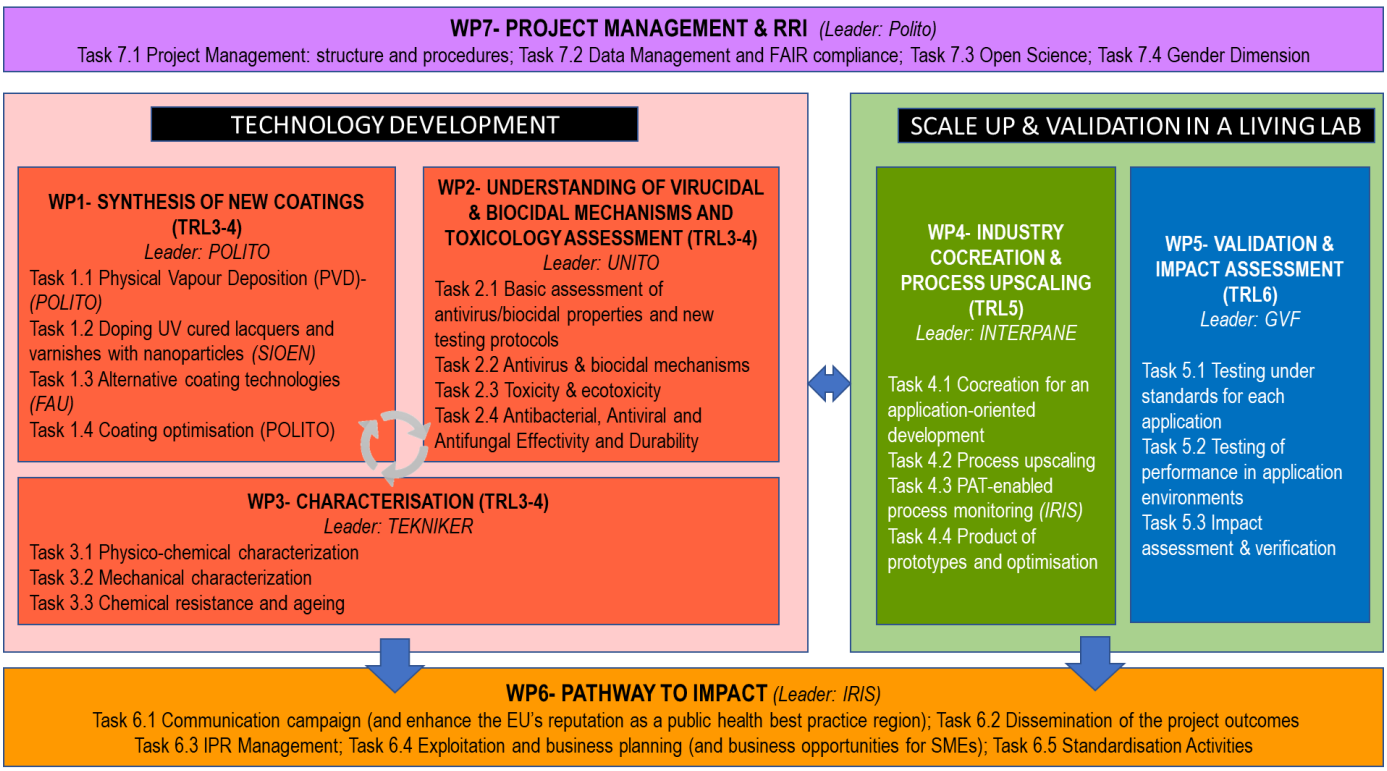


Table 3.1g: 'Subcontracting costs' items

No subcontracting is envisaged.

Table 3.1h: 'Purchase costs' items (travel and subsistence, equipment and other goods, works and services) and 'Other costs categories' items (e.g. internally invoiced goods and services)

1 POLITO	**Cost (€)**	**Justification**
Travel	21300	Travel fund to cover travel & subsistence for: 2 people @ 1 presential project meeting per year (supplemented with virtual meetings)- about €9K; 2 people @ 1 or more International conference/exhibition per year (EU or outside EU)- about €12K;
Equipment	4000	Depreciation cost of equipment for the coating and prototype characterization
Other goods, works and services	72600	Targets for co-sputtering, technical gases, laboratory consumables, chemicals, consumables for antibacterial tests (22000 €); Registration fees for participation in 6 conferences (average value 600 €: 3600 €); open access publication (5000 €); spare parts of sputtering equipment (2000 €); IP cost (2000 €); audit (5000 €); service contract for a technician specialized in co-sputtered coating (30000 €)
Total	**97900**	
2 TEKNIKER	**Cost (€)**	**Justification**
Travel	15000	Travel fund to cover travel & subsistence for: 2 people @ 1 presential project meeting per year (supplemented with virtual meetings)- €6K; 2 people @ 1 International conference/exhibition per year (EU or outside EU)- €6K; attendance at national events & travel to partner sites & labs (€3k)
Equipment		
Other goods, works and services	107000	WP1: Cu, Cr, Ti, Zr targets for PVD coatings (19k€) & Magnetron sputtering (45k€) for Cu evaporation to be incorporated in KSA Cathodic arc existing equipment. It is necessary to carry out the combination of Cathodic arc Nitride evaporation and Cu magnetron sputtering to allow better control on the Cu % in the formulations; WP2 (bio, toxi, biocide test): consumables for toxicity tests (different kits, daphnia, algae, v.fischeri) 5k€, chemical reactives (2k€), kits for biocide assessment (6k€); WP3 (ageing): consumables for mechanical-tribological-ageing lab (5k€), samples preparation (4k€), holders for mechanical/ageing configurations (6k€); WP4 ancillary tools 6 substrate holders for upscaling (5k€) sol-gel chemicals & precursors (5k€); WP5 (LCA): software license and database access (2k€); WP7: audit certificate (3k€)
Total	**122000**	
3 UNITO	**Cost (€)**	**Justification**
Travel	15000	Travel fund to cover travel & subsistence for: 2 people @ 1 presential project meeting per year (supplemented with virtual meetings)- €6K; 2 people @ 1 International conference/exhibition per year (EU or outside EU)- €6K; attendance at national events & travel to partner sites & labs (€3k)

Equipment	25000	Depreciation cost of Lab equipment for the establishment of a supplementary dedicated supply chain: CO2 incubator for cell line culture, virus production and antiviral experiments incubation; -80°C refrigerator for viral stock and cell lines conservation; biosafety cabinet level II for the execution of the antiviral assays; -20°C refrigerator for stock of cell culture reagents; supercentrifuge for the production of viral stocks; multimode microplate reader for cytotoxicity assays
Other goods and services	124300	Laboratory consumables (flasks, pipets, 96 well plates, gloves, ultracentrifuge tubes, scrapers, 50ml and 15ml tubes, eppendorf...) (€ 85000), Reagents for cell culture (cell culture medium, fetal bovine serum, antibiotics, trypsin, glutamine, aminoacids) (€ 8000), Feline calicivirus and cell line for virus production and antiviral assays (the other 5 viruses and cell lines are still available in the UNITO Lab)(€ 1300), antibodies for immunocitochemistry assays (€ 3000), MTT kit for cytotoxicity assays (€ 5000), reagents for TEM analysis (€ 8000), Instrument Maintenance (€ 10000), audit certificate (€ 4000)
Total	**164300**	
4 FAU	**Cost (€)**	**Justification**
Travel	9000	Travel fund to cover travel & subsistence for: 2 people @ 1 presential project meeting per year (supplemented with virtual meetings)- €6K; 1 people @ 1 International conference/exhibition per year (EU or outside EU)- €2K; attendance at national events & travel to partner sites & labs (€1k)
Equipment		
Other goods, works and services	28600	Consumables (substrates and solvents for EPD and sol-gel method); Open Science costs (4k); Registration fees for participation in 6 conferences (average value €600x6= €3600)
Total	**37600**	
5 GVF	**Cost (€)**	**Justification**
Travel	10000	Travel fund to cover travel & subsistence for: 2 people @ 1 presential project meeting per year (supplemented with virtual meetings)- €4K; 2 people @ 1 International conference/exhibition per year (EU or outside EU)- €4K; attendance at national events & travel to partner sites & labs (€2k)
Equipment		
Other goods, works and services	40000	Sintered fibers, wire mesh, Spunbound/Meltblown PP textile, special alloys any raw materials and consumables for filters manufacturing.
Total	**50000**	
6 SIOEN	**Cost (€)**	**Justification**
Travel	15000	Travel fund to cover travel & subsistence for: 2 people @ 1 presential project meeting per year (supplemented with virtual meetings)- €6K; 2 people @ 1 International conference/exhibition per year (EU or outside EU)- €6K; attendance at national events & travel to partner sites & labs (€3k)
Equipment	6000	Light cabinet with different wavelengths (depreciated cost)
Other goods, works and services	105000	Lab consumables (10k€); Glassware (5k€); Extra tests with ceramic beads at Netzsch (8k€); external sourcing NPs (5k€): wear testing (12k); excimer test (15k) Standardized assay services for product validation (TRL5) (47k); audit certificate (3k)
Total	**126000**	
7 PRIREV	**Cost (€)**	**Justification**
Travel	15000	Travel fund to cover travel & subsistence for: 2 people @ 1 presential project meeting per year (supplemented with virtual meetings)- €6K; 2 people @ 1 International conference/exhibition per year (EU or outside EU)- €6K; attendance at national events & travel to partner sites & labs (€3k)
Equipment	68800	Calotest: 7 500€; Microhardness Tester: 27 000€; LIS (Linear Ion Source): 31 875€
Other goods, works and services	144375	Sputtering rotatable targets (length 1334mm, diameter 70mm; materials: Ag, Cr, Ti, Zr)- 35k; Cathodic Arc Targets (diameter 100mm; Cr, Ti, Zr, Cr-Ag, Ti-Ag)- 20k, Gases (Ar, N2, CH4, C2H2, O2)-6k; consumables to clean surfaces pre-PVD deposition (solvents (ethanol, acetone, propanol), alkaline ultrasonic cleaning products, shot blasting raw materials)-12k ; chemical decoating products-6k ; Substrate holders- 12k; Consumables for characterization equipments (reagents, silicon wafers, abrasives)-3k. External characterization services provided by laboratories equipped with advanced and more suitable instrumentation for the evaluation of coating properties 50k€
Total	**228175**	
8 IRIS	**Cost (€)**	**Justification**
Travel	15000	Travel fund to cover travel & subsistence for: 2 people @ 1 presential project meeting per year (supplemented with virtual meetings)- €6K; 2 people @ 1 International conference/exhibition per year (EU or outside EU)- €6K; attendance at national events & travel to partner sites & labs (€3k)
Equipment		
Other goods, works and services	103000	Analysers - 50 k; QT SOFTWARE LICENSE FOR THE DEVELOPMENT OF THE APPLICATIONS USED IN THE MONITORING PROTOTYPE - 10K€ Optomechanical materials - 10k; computing for prototype (5k); software licenses (4k); Audit certificate (4k); IP cost (Patent lawyer & patent application) - 8k; subscription to events & exhibition space - 6k; Data Management Cost (6k)
Total	**118000**	

9 INTERPANE	Cost (€)	Justification
Travel	15000	Travel fund to cover travel & subsistence for: 2 people @ 1 presential project meeting per year (supplemented with virtual meetings)- €6K; 2 people @ 1 International conference/exhibition per year (EU or outside EU)- €6K; attendance at national events & travel to partner sites & labs (€3k)
Equipment	50000	Depreciation of plant coating equipment
Other goods, works and services	405000	Hardware costs: cathode co-sputtering (cathode lid, TMPs, MFCs, control cards, PLC and automation, cooling system/chiller, feedthrough, shields, cables, pipes and tubes, valves, foreline line modification, end blocks (in case of rotable cathode), etc.) - €297k; power supply (PSU + power cable + etc.) - €25k; silver target - €60k; silicum target- €10k; Sensors - €10k; Audit Certificate -3k
Total	**470000**	
10 UNE	**Cost (€)**	**Justification**
Travel	7000	Travel to consortium meetings and travel to attend dissemination events
Equipment		
Other goods, works and services	4000	Free access to standards for consortium partners; fee for registration of CWA(s) at CEN
Total	**11000**	

Table 3.1i: 'Other costs categories' items (e.g. internally invoiced goods and services)
No partners have budgeted costs under this category.

3.2 Capacity of participants and consortium as a whole

Consortium description, complementarity, access to critical equipment and industry involvement

The NanoBloc® consortium unites a world-class team of 10 partners from 5 European countries (fig. 10)- Italy, Spain, Portugal, Germany, Belgium. Pooling together **complementary expertise and competencies** from industrial leaders across a complete **European industry value chain** and scientifically excellent academic/research universities and institutes, the NanoBloc® consortium converges the different disciplines needed for successfully cocreating, developing and validating ground-breaking reliable, high performance and scientifically validated antiviral, antibacterial and antifungal nanocoating technologies. The NanoBloc® team has the ambition through research, development and innovation to establish European industrial leadership in the field of high performance nanocoatings that can help Europe protect its citizens, its healthcare system and its economy from the devastating effects of infectious illnesses, such as the current Covid-19 pandemic

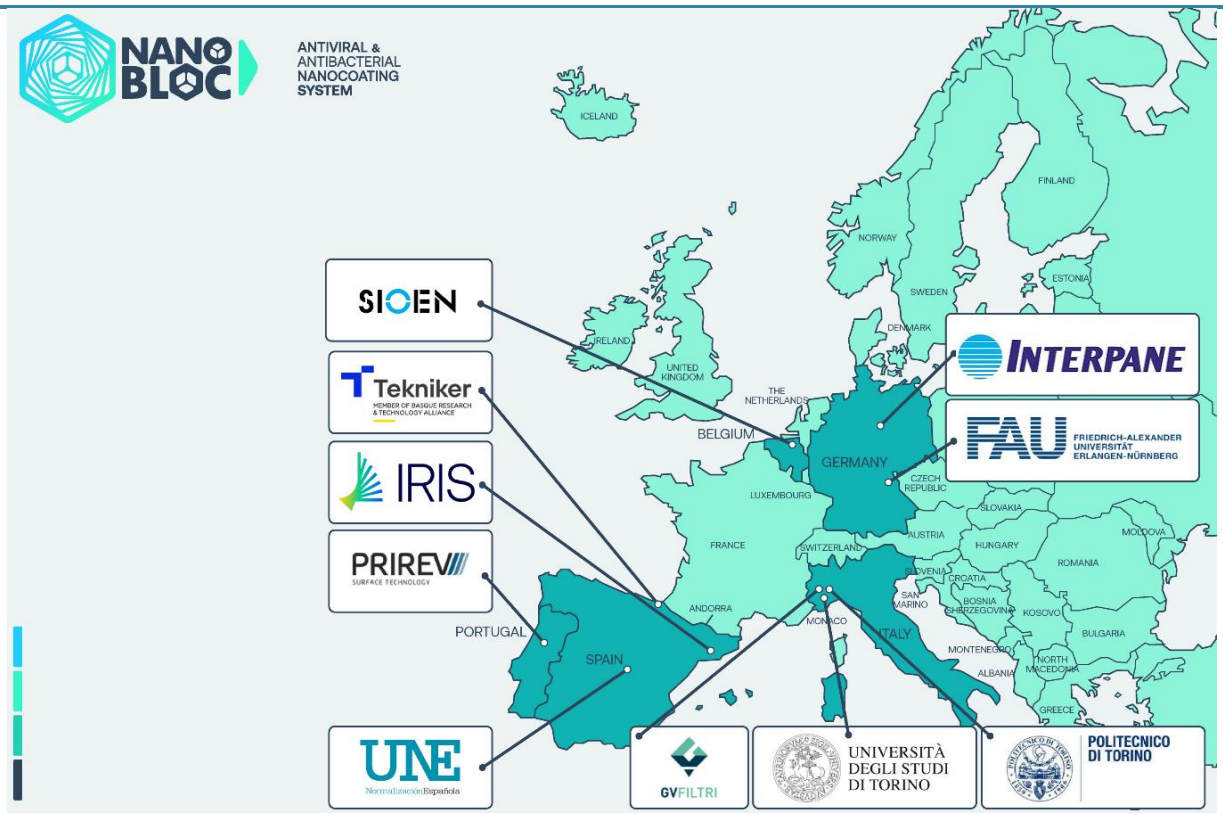


Figure 10- The Nanobloc consortium from 5 European countries

- RF co-sputtering of Ag/silica (and other metal/ceramics) and their characterization (POLITO)
- EPD and sol-gel (FAU)

- HiPIMS, Cathodic arc, and sputtering PVD coatings, sol-gel coatings development and up-scaling and tribological/durability characterization (TEKNIKER)
- Cathodic arc evaporation & sputtering (PRIREV)
- UV-lacquers coated technical textiles (SIOEN)
- Antimicrobial/antiviral testing (UNITO)
- Coating equipment upscale (INTERPANE)
- Filtration systems, production and commercialization (GVF)
- Photonics & Artificial Intelligence key enabling technologies for digitization of industrial processes (IRIS).
- Standardization activities (UNE)

This is an **integrative consortium** that includes, four world class universities and institutes (POLITO, UNITO, TEKNIKER, FAU), who are **leaders in their respective scientific and technological fields**, in addition to two large enterprises (INTERPANE, SIOEN) and 3 SMEs (IRIS, GVF, PRIREV) and a standardization body (UNE), who together will provide a **technology upscaling and market pathway** for the projects results. The NanoBloc® team of partners unites **the necessary infrastructure, knowledge, expertise, and industry value chain to succeed** in delivering on the project's objectives to realise the expected impacts of the topic 'HORIZON-CL4-2021-RESILIENCE-01-20: Antimicrobial, Antiviral, and Antifungal Nanocoatings'.

Figure 11- Industrial Value Chain

As a condition to generate growth, NanoBloc® is an **industry-oriented project**. The **industrial/commercial involvement** spans a complete value chain of industrial actors (figure 11), 3 of which are SMEs (IRIS, PRIREV, GVF), is imperative for allowing us to implement a research-industry cocreation in our project (according to the methodology described in Section 1.2). **Each industry participant will be able to grow their businesses** and improve their competitiveness in line with their position on the value chain. **This involvement ensures a technology pathway for the project upscaling (INTERPANE, IRIS, PRIREV), and a route to market (GVF, SIEON and PRIREV**); a complete value chain for the exploitation, post-project scale up and commercialisation of the projects results. In short, **the industry participants have been instrumental in helping to shape this proposal, and NanoBloc® is considered a highly strategic for assisting them in gaining a competitive edge and in being a pioneering group in driving forward new antiviral, antibacterial & antifungal nanocoating innovations of high socio-economic & strategic impact**.

In addition, the consortium unites a strong understanding of **gender issues and inclusiveness**, and also has an **equitable Gender dimension**, whereby there is **a strong participation of female researchers**, technologists, industrialists, business experts and entrepreneurs, as well as policy influencers. Moreover, the majority of our partners either have a formal Gender Equality Plan (GEP) in place or are in the process of implementing one, and all partners have equality opportunities policies in relation to recruitment and internal promotions. Moreover, our **Project Coordinator (Prof. Monica Ferraris),** is an accomplished, award-winning female scientist- she is a Full Professor of Science and Technology of Materials at POLITO, with over 200 hundred peer-reviewed journal publications and 14 patents. She is also featured in **Successful Women Ceramic and Glass Scientists and Engineers: 100 Inspirational Profiles**[69]. *She acted as Project Coordinator or Principal Investigator of 27 National and EU projects on ceramics, glasses and composite, from 1992 to present: among them, 17 are EU projects, 8 of them as Project Coordinator*. In relation to open science practices, our research partners have considerable expertise in this area, and they are active publishers in open science journals, and also committed to seeing scientific results

[69] Lynnette D. Madsen. 2015 The American Ceramic Society and John Wiley & Sons, Inc. Published 2015 by John Wiley & Sons, Inc (page 175-179)

being exploited by industry for building competitiveness, growth, jobs, resilience and societal impact.

How each member contributes to the project with a valid role and with the necessary resources

The table below summarises the core role of each of the partners in the project.

Table 22- Role of each partner

Partner	Role
POLITO	**Coordinate Project.** WP1 & WP7 leader. Participant in WP2, WP3, WP4, WP5, WP6. **Core role**: Co-sputtering provider of Ag/silica coatings (and other compositions). Contributing background in novel technique for the deposition of a nanostructured coating with virucidal, antimicrobial and antifungal properties via co-sputtering technique, patented by POLITO (PCT/IB2018/057639). Also perform routine characterization on own samples and partners samples upon request. Leader of Open Science and Gendered Innovation approach.
TEK-NIKER	WP3 leader. Participant in WP1, WP2, WP4, WP5, WP6, WP7. **Core role:** Contribute vast experience on tribological, wear and ageing (UV, Corrosion, oxidation, etc.) behaviour & characterisation for different applications, on antibacterial performance & marine ecotoxicity testing, on coatings with biocide behaviour and durability (medical implants, anti-fouling) and experience on Life Cycle Analysis (LCA). Leadership of coatings characterization (WP3); contribute to R&D on new PVD coatings containing Cu (WP1) using different subtechnologies & combinations (cathodic, arc, sputtering, HIPIMS); antibacterial testing & ecotoxicity tests (WP2); collaborate on up-scaling of PVD batch processing for high traffic objects & Sol-gel upscaling (WP4); Validation in relevant environments & LCA (WP5). Role of Scientific Manager to support interdisciplinary mgt.
UNITO	WP2 leader: Participant in WP6, WP7. **Core role:** Contribute their expertise in the study of the mechanism of action of new antiviral molecules and in the study of molecule cytotoxicity to lead the evaluation of the antiviral properties of different coatings produced in the project and the study of their mechanism of antiviral action and cytotoxicity. They will also evaluate the influence of ageing processes on the antiviral properties of the coatings.
FAU	Participant in WP1, WP3, WP6, WP7. **Core role:** Thanks for their world-class experience in the field of bioactive materials, coatings and electrophoretic deposition method, FAU will lead the development and characterisation of antiviral and antibacterial nanocoatings by EPD and sol-gel method, & the incorporation of bio-based materials.
GVF	WP5 leader. Participant in WP1, WP2, WP3, WP4, WP6, WP7. **Core role:** As experts in the design, development and production of industrial filtration elements, equipment and systems, GVF will work on the effective application of a virucidal/antimicrobial/ antifungal coating on filtration systems for improving product quality and reliability (WP4). In addition to leading WP5, they themselves will test NanoBloc® on air filtering materials following standards and monitoring performance, and measure the impact and success to demonstrate the new nanocoating technologies in relevant working environments. Plan post-project pathway to market (WP6).
SIOEN	Participant in WP1, WP3, WP4, WP5, WP6, WP7. **Core role:** Contribute their experience in the formulation of different nanoparticle dispersions for technical textile coatings & radiation-curable lacquers and varnishes, and lead the work in relation to the doping UV cured lacquers & varnishes with nanoparticles to obtain self-disinfecting properties (Task 1.2). Test & demonstrate the technology (WP5) on coated textiles (incl. protective clothing, mattresses, and mouth masks), lacquers, and varnishes. Plan post-project pathway to market (WP6).
PRIREV	Participant in WP1, WP4, WP5, WP6, WP7. **Core role:** Contribute their 20+ years of PVD coating expertise to the up scaling of the deposition process for antibacterial/antiviral coatings in line with their R&D roadmap to develop innovative new products such as the recent development of AB20- a new antibacterial coating, inox colour, for stainless steel substrates with antibacterial activity certified by Hohenstein Institute. Main goal is to now develop coatings in other colours with antibacterial/antiviral properties, for different substrates (brass, zamak, polymers) compliant with the envisaged applications of high traffic objects, and plan pathway to market.
IRIS	WP6 leader. Participant in WP4, WP7. **Core role**: As an advanced engineering company specialising in PAT (Process Analytical Technology) and Artificial Intelligence real-time process monitoring solutions for Industry 4.0, lead the work in T4.3 in relation to measuring the Critical Process Parameters and the target Critical Quality Attributes of the selected processes, and building Digital Twins based on the data historian for improved process understanding (based on Quality-by-Design approach) and predictability. They will also occupy the role of Data Managers for the project. Lead implementation of the Cocreation Cycle in the project (T4.1). Lead IP Management (T6.3), Exploitation & Business Planning (T6.4) for planning timely routes to market.
INTERPANE	WP4 Leader. Participant in WP5, WP6, WP7. **Core role:** Upscaling of the R2R processes: engineering studies, hardware modifications for co-sputtering Ag nanoclusters in a silica matrix. Install & commission the modified hardware in R2R demo unit & co-sputtering coating deposition on substrates selected by end users. Plan post-project equipment TRL7-9 scale up and market deployment.
UNE	Participant in WP6, WP7. **Core role:** Leading Standardization activities and procedures

Critical infrastructure. Major items include: **POLITO**- fully equipped laboratories for research on glasses, glass-ceramics, ceramics, composites, surface functionalization and nanostructured coatings; **UNITO**-Laboratory of Molecular Virology and Antiviral Research; access to San Luigo Hospital (that hosts UNITOs lab) for validation work; **TEKNIKER**-advanced national laboratory in PVD technology, surface chemistry laboratory for Sol-Gel coatings, clean room with UV photolithography equipment, plasma etching and electroplating systems, equipment for Hot embossing / Nanoimprint and R2RNIL machine, etc., & Tribology laboratory; **FAU**- facilities for processing

and characterisation of biomaterials, including mesoporous materials and composites, powder technology and biomaterial preparation and characterisation suitable chemistry laboratories are available for sol-gel; **INTERPANE-** Roll-to-roll PVD & PECVD coater for maximum width of 1.6 meter; Material characterisation equipment; **PRIREV**- PVD deposition chambers (incl. Cathodic arc and sputtering technologies). **SIOEN**- UV & Led-curing equipment, Extrusion coating lines, calandering lines, overcoat lines, direct coating lines, transfer coating lines, dip coating lines, lamination coating lines etc. Hot Xenon UV weathering equipment; **GVF**- Pleating machine, Tig welding machine, Microplasma welding machine, Rolling machine; **IRIS**- platform of IR (SWIR) analysers.

Adequate resources to fulfil their role, each partner is experienced in building budgets and in planning forward required resources and there is a logical budget distribution among the partners. Each has mobilised the necessary **personnel resources/person months** (*Table 3.1f: Summary of staff effort*) that reflect the realistic staff effort required to deliver the project objectives. Personnel budget is based on actual costs for researchers, engineering, developers and other professionals for each partner organisation. These costs are in line with average staff costs within the sectors and reflect differences in cost of living between the participating EU regions, whereby staff costs in Northern Europe (namely, Belgium and Germany) are higher versus Southern Europe (namely, Portugal, Spain, Italy) and this difference is particularly pronounced for the private sector. **Equipment costs** included in the project are for **depreciation only**, and as relevant partners have budgeted and itemised this equipment in *Table 3.1h: 'Purchase costs' items (travel and subsistence, equipment and other goods, works and services)*. A reasonable budget has been allocated under **travel and subsistence** (*a flat travel fund of €15k has been for each partner*) to allow for mobility of developers and industrialists between sites, which is important for knowledge exchange and building excellence among multidisciplinary and international teams, as well as for practical on-site testing across partners. *(Importantly, our intention is to facilitate and enable a high level of virtual engagement among the project team, to supplement the annual presential/onsite project meetings (as allowed). In this way we will have high resilience to any future travel restrictions, confinements and lockdowns. Moreover, promoting virtual interactions and engagements will be positive for our contribution as a project to carbon emissions reduction*). No partners have budgeted for subcontracting costs, as they each have access to the internal resources and knowledge to perform the work. Partners have budgeted for **diverse goods and services**, e.g. funds for Laboratory and Prototyping consumables, publishing in Open Journals, Patenting fees (attorney and application costs), Subscription to Conferences and Events, Open Data Storage, among others (as detailed and itemised in Table 3.1h). Noteworthy is €350k by INTERPANE materials for the scale-up **cathode co-sputtering** cost. This is an important cost that covers the price for the PVD lid (material + manufacturing) which is used to close the R2R and fix the cathode targets and other parts. Other parts are mainly: MFCs + control system related; TMPs + control system related; Electrical feedthrough; Water cooling system for TMPs and cathodes; Shields for cathodes and R2R; Various valves (inline, venting, bellows, etc.); PLC and automation system (e.g. Beckhoff); Cables and related; Stainless steel pipes and plastic tubes; End blocks (in case of rotatable cathode, etc.). This budget item has been given careful assessment collectively by the consortium and it is considered necessary and relevant for delivering a **high-end system for NanoBloc®**. In short, the partners have worked very closely together to codesign the workplan and its associated budget. All partners are bringing a clear value to the project, are performing a synergising and complimentary role, and with the necessary resource to be able to implement this role and each partner is highly motivated with a core role category as either a **knowledge creator** (POLITO, UNITO, TEKNIKER, FAU), a **technology upscaler** (INTERPANE, FAU, PRIREV) or as an **innovation adopter** (GVF, SIOEN), and a standardization facilitator (UNE).

Other countries and international organisations: Not applicable

4. Ethics Self-Assessment

In relation to the Ethics Self Assessment table, while the overall 'Ethics Opinion' in the Ethics Summary Report concluded 'Ethics clearance (the proposal is 'ethics ready')', its 'Analysis of the ethical dimension' concluded the following:

'The ethics issues relating to Section 4 - Personal Data, Section 7 - Environment, Health & Safety (nanomaterials) and Section 8 - Artificial Intelligence are not clearly identified in the self-assessment form, and the beneficiaries should consider any potential ethics issues under these sections. In addition, it is not clear whether the mammalian cells used in task 2.3 will be human cells which raises ethical issues.'

The NANOBLOC consortium confirms that compliance with the ethical principles and the applicable EU, international, and national laws for the ethics issues identified in the Ethics Summary Report any ethics issues that may emerge in the course of the grant, will be ensured, and for any applicable ethics issue, the guidance provided in the European Commission Ethics Self-Assessment Guidelines will be rigorously followed. In relation to the mammalian cells used in task 2.3, we clarify that the term "mammalian cells" includes human cells that we will use

in the project. However, it must be pointed out that only widely used (commercial) human cell lines will be used. They are purchased from cells banks, such as ATCC (the world's largest and most extensive product catalogue of human and animal cell lines for research purposes). We will not derive any cells from human donors during the project.